Evaluating Machine Learning Models

How to measure, compare and trust models before and after they reach real users

MEASURE

Splits, metrics, calibration

PROVE

Statistics, slices, robustness

TRUST

Fairness, online tests, monitoring

GETTING STARTED

Why evaluation deserves a whole tutorial



A model is only as trustworthy as the evidence behind it. Evaluation is how that evidence is created.

The problem

Teams often report one impressive number that hides leakage, bias or a weak baseline.

The cost

Poorly evaluated models fail quietly in production, harming users and wasting investment.

The goal

Give you a rigorous, practical method for measuring models honestly from first split to live traffic.

GETTING STARTED

1 Data scientists

Who choose metrics, run experiments and must defend their results.

2 ML engineers

Who build evaluation pipelines, quality gates and monitoring.

3 Analysts

Who design A/B tests and interpret experiment results.

4 Product managers

Who must judge whether a model is good enough to launch.

5 Reviewers and auditors

Who assess fairness, robustness and compliance evidence.

6 Students

Who want to move beyond accuracy to professional evaluation practice.

GETTING STARTED

What you should already know

- What training and prediction mean
- The difference between classification and regression
- Basic Python and a library such as sklearn
- Reading simple charts and tables
- The idea of averages and percentages
- Helpful but optional: basic probability
- Helpful but optional: experiment tracking tools
- Helpful but optional: SQL for slicing data

GETTING STARTED

Learning outcomes

1 **Design sound splits**

Create training, validation and test sets that reflect real use and avoid leakage.

2 **Choose the right metric**

Match classification, regression, ranking and forecasting metrics to business goals.

3 **Judge probabilities**

Evaluate thresholds, calibration and the cost of different errors.

4 **Prove differences**

Use confidence intervals and significance testing to compare models honestly.

5 **Look deeper**

Run error analysis, slicing, fairness, robustness and uncertainty checks.

6 **Evaluate live**

Plan A/B tests and production monitoring that confirm real value.

GETTING STARTED

Roadmap of the twenty one sections

No.	Section	No.	Section
1	Foundations of evaluation	12	Evaluating LLMs and generative models
2	Data splitting and validation	13	Statistical rigour
3	Data leakage	14	Error analysis and slicing
4	Classification metrics	15	Fairness evaluation
5	Thresholds, probabilities, calibration	16	Robustness and uncertainty
6	Regression metrics	17	Interpretability checks
7	Imbalanced data	18	Online evaluation and A/B tests
8	Ranking and recommendation	19	Evaluation in production
9	Forecasting	20	End to end case study
10	Clustering and anomalies	21	Checklists, glossary and quiz
11	Language, vision and speech		



Each section opens with a divider and closes with a recap, an exercise and a preview of what comes next.

GETTING STARTED

How the slides are designed to be read

Definition slides	Explain one term with an example and a memory hook.
Comparison slides	Place two options side by side and end with a verdict.
Flow slides	Show a process as numbered boxes connected by arrows.
Practice slides	Code, checklists, pitfalls and knowledge checks.
Recap slides	Summarise, set an exercise and preview the next section.



Reading guide

Blue boxes carry core ideas. Orange boxes warn about traps. Yellow sticky notes hold tips worth remembering. Dark boxes give verdicts and next steps.

SECTION 1

01

Foundations of evaluation

What we are really measuring, why, and the mindset that keeps evaluation honest.

IN THIS SECTION

- What evaluation means
- Offline versus online
- Generalisation
- Overfitting and underfitting
- Baselines
- Linking metrics to value

SECTION 1 · FOUNDATIONS

What is model evaluation?

DEFINITION

Model evaluation is the systematic process of estimating how well a model will perform on the data and decisions it will face in real use, and whether that performance is good enough.

EXAMPLE

Before replacing a manual triage process, a hospital team checks how often the model ranks urgent cases correctly on recent records it has never seen, compared with current practice.

REMEMBER

Evaluation estimates future performance. It never just describes the past.

SECTION 1 · FOUNDATIONS

Why we evaluate

1 Select
Choose between candidate models, features and settings.

2 Decide
Judge whether a model is good enough to launch at all.

3 Communicate
Give stakeholders honest evidence about benefits and risks.

4 Protect
Detect harm to users or groups before it reaches them.

5 Improve
Find specific weaknesses to fix in the next iteration.

6 Monitor
Confirm performance holds once the model is live.

SECTION 1 · FOUNDATIONS

Offline versus online evaluation

OFFLINE

Measured on stored data

- Fast, cheap and repeatable
- Uses historical labels
- No risk to real users
- May not reflect behaviour change caused by the model

ONLINE

Measured with real users

- Captures true business impact
- Slower and more expensive
- Carries some user risk
- Needs careful experiment design

VERDICT Use offline evaluation to filter and refine candidates, then confirm the winner online. Neither replaces the other.

SECTION 1 · FOUNDATIONS

Generalisation

DEFINITION

Generalisation is a model's ability to perform well on new data drawn from the same situation it will be used in, not just on the examples it learned from.

EXAMPLE

A spam filter that scores perfectly on its training emails but misses half of next week's spam has memorised rather than generalised.

REMEMBER

Every evaluation decision exists to estimate generalisation honestly.

SECTION 1 · FOUNDATIONS

Overfitting and underfitting



Good fit

Low error on training data and on new data. The goal.

Overfitting

Low training error but high error on new data. The model memorised noise.

Leakage warning

Suspiciously low error everywhere. Check for leaked information.

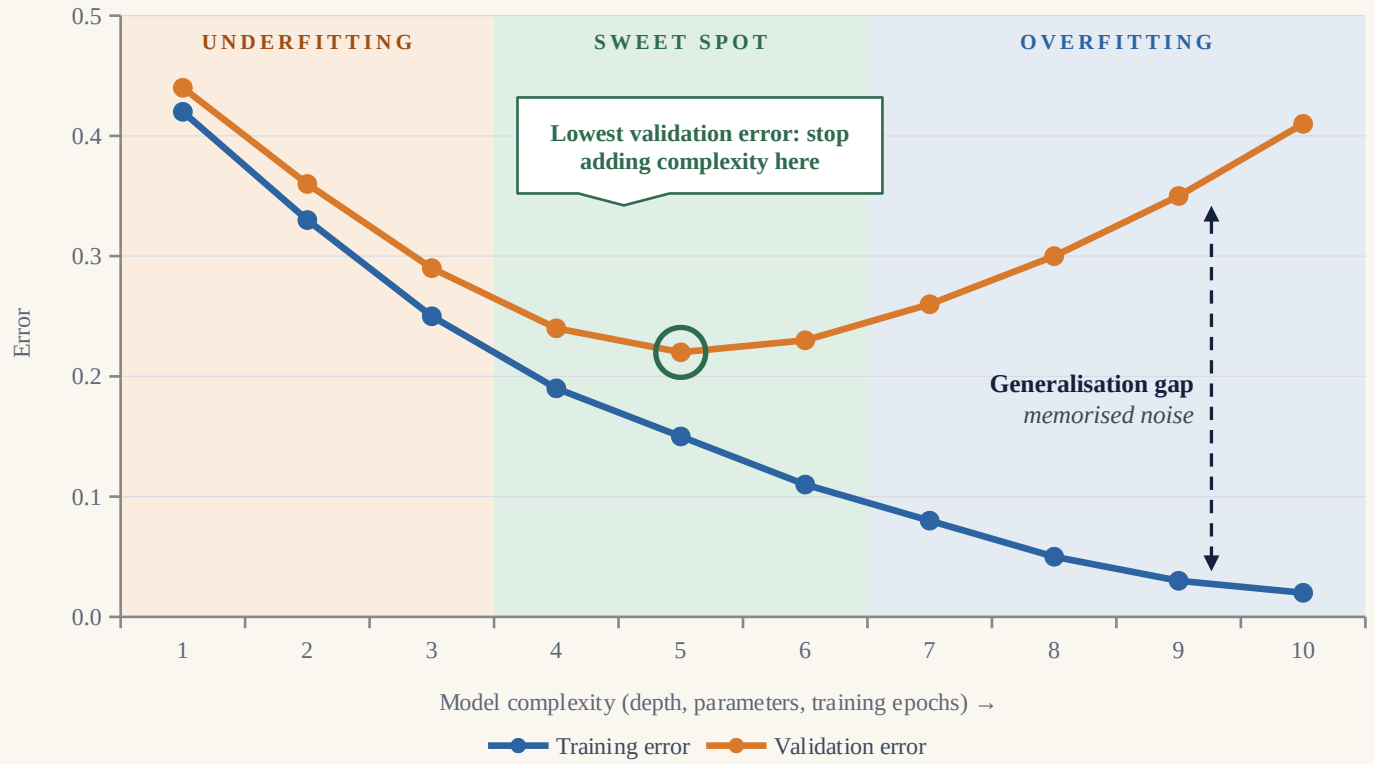
Underfitting

High error on both. The model is too simple or features are weak.

Vertical axis: Error on training data (low at top)

Horizontal axis: Error on new data (low at left)

Overfitting, drawn as two error curves



THE EXAM ANALOGY

UNDERFITTING

Skimmed the notes

Fails the practice papers and the real exam. The model is too simple for the pattern.

GOOD FIT

Understood the ideas

Scores well on practice papers and on questions never seen before.

OVERFITTING

Memorised past papers

Perfect on familiar questions, lost on new ones. The training score is lying.

! Watch the validation curve, not the training curve. Once it turns upward, extra complexity is buying memorisation, not skill.

SECTION 1 · FOUNDATIONS

Always start with a baseline

A metric means nothing in isolation. A baseline shows what performance is achievable with no model or a very simple one.

Trivial baselines

Predict the most common class, the average value or yesterday’s value.

Simple models

Logistic regression, a small decision tree or a rule written by an expert.

Current process

The existing human or rules based decision, which the model must beat to be worth it.

SECTION 1 · FOUNDATIONS

From business goal to evaluation metric

01

Goal

State the outcome the business wants, such as fewer missed frauds.

02

Decision

Describe the action the prediction triggers.

03

Costs

Estimate the cost of each type of mistake.

04

Metric

Pick metrics that reflect those costs and actions.

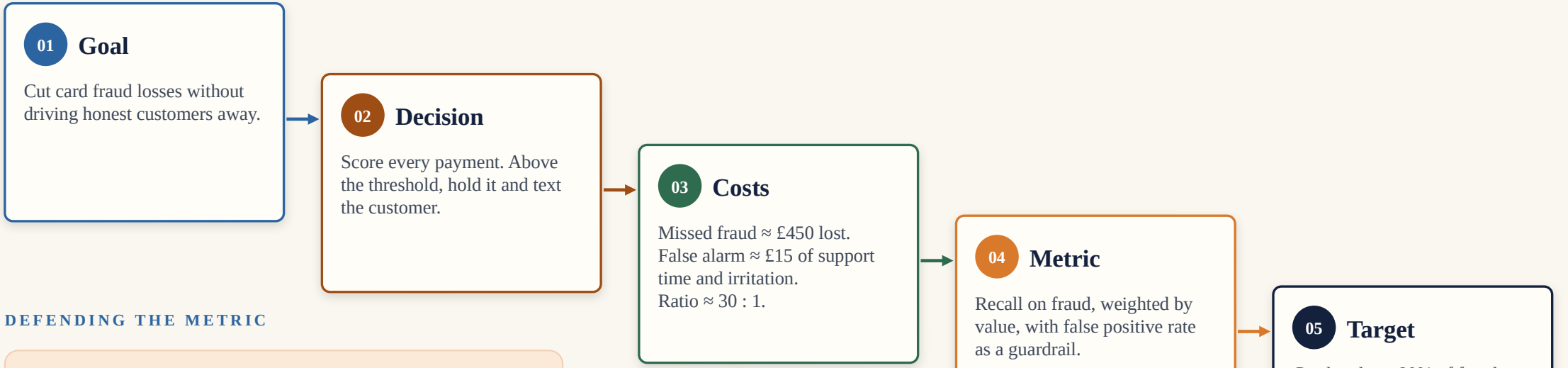
05

Target

Agree the level that counts as good enough.

If you cannot explain why a metric matters to the business, it is probably the wrong metric.

Goal to metric, worked through for a fraud model



DEFENDING THE METRIC

Stakeholder: Why report recall instead of accuracy? Accuracy is 99.6%.

You: Fraud is under 1% of payments, so 'approve everything' also scores 99%. A missed fraud costs about 30 times a false alarm, so what we must maximise is the share of fraud we catch.

! Put the cost ratio on the slide. That one line justifies the metric, the threshold and the target.

Levels of evaluation

Technical	Does the model fit the data? Loss and training curves.
Predictive	How accurate is it on unseen data? Core metrics.
Behavioural	Does it behave sensibly across slices and edge cases?
Responsible	Is it fair, robust, calibrated and explainable?
Business	Does it improve the outcome that matters, measured live?



Climb the ladder

Good teams evaluate at every level. A model can top the leaderboard at the lowest level while failing badly at the highest.

SECTION 1 · FOUNDATIONS

Foundational mistakes

PITFALL	BETTER PRACTICE
⚠ Reporting one headline metric only.	✓ Report several metrics, a baseline and confidence intervals.
⚠ Choosing the metric after seeing results.	✓ Agree metrics and targets before running experiments.
⚠ Evaluating on data the model has seen.	✓ Hold out data that stays untouched until final testing.
⚠ Ignoring the cost of different errors.	✓ Weigh mistakes by their real world consequences.

SECTION 1 · FOUNDATIONS

Knowledge check

? A model reaches 99 percent accuracy at detecting a disease that affects 1 percent of patients. Is it impressive?

MODEL ANSWER

Not necessarily. A model that always predicts no disease also scores 99 percent accuracy while finding nobody. You must compare against this baseline and look at how many real cases it detects and how many alarms are false.

WHY IT MATTERS

It shows why baselines and the right metric matter more than a big number.

SECTION 1 · FOUNDATIONS

Section recap

KEY TAKEAWAYS

- Evaluation estimates performance on future, unseen data
- Offline evaluation filters, online evaluation confirms
- Watch for overfitting, underfitting and leakage
- Always compare against simple baselines
- Derive metrics from business goals and error costs
- Evaluate at technical, predictive, responsible and business levels

TRY THIS

For a model you know, write its business goal, the decision it drives, the cost of each error and the metric you would choose.

NEXT UP

Section 2: splitting data and validating models correctly.

SECTION 2

02

Data splitting and validation

Designing the data you test on so it honestly represents the data you will face.

IN THIS SECTION

- Training, validation, test
- Random and stratified splits
- Time based splits
- Group splits
- Cross validation
- Choosing split sizes

SECTION 2 · SPLITTING AND VALIDATION

Three sets, three jobs

Training set	Used to fit model parameters. The model sees it many times.
Validation set	Used to compare models, tune settings and choose thresholds.
Test set	Used once to estimate final performance on unseen data.
Optional extra	A fresh, later sample or external dataset to test transfer.

Golden rule

The test set is opened once, at the end. Every time you look at it and adjust, it becomes a validation set and stops being an honest estimate.

Three sets, explained as one student's school year



TRAINING

The textbook exercises

Used to fit the parameters. The model works through these again and again.

SEEN THOUSANDS OF TIMES

VALIDATION

The mock exams

Used to compare models, tune settings and pick thresholds. Every choice leaks a little of it into the model.

SEEN MANY TIMES

TEST

The sealed final exam

Opened once, at the very end, to estimate how the model will do on data it has never met.

OPENED ONCE

OPTIONAL EXTRA

The external examiner

A later or outside sample that checks the result still holds somewhere new.

ONCE, LATER

RULE Each time a test score changes a decision, the test set quietly becomes a validation set. When that happens, collect a fresh one.

Proportions are typical, not fixed: what matters is enough examples of the rarest important case in each set.

SECTION 2 · SPLITTING AND VALIDATION

Random versus stratified splits

RANDOM

Shuffle and cut

- Each row assigned at random
- Fine for large, balanced datasets
- Rare classes may vanish from small sets
- Simple and widely used

STRATIFIED

Preserve class proportions

- Each set keeps the same class balance
- Essential for imbalanced problems
- More stable metric estimates
- Easy option in most libraries

VERDICT Stratify by the target for classification by default, especially when any class is rare.

Time based splits

DEFINITION

A time based split trains on earlier data and tests on later data, so evaluation mimics how the model will be used: predicting a future it has not seen.

EXAMPLE

A demand model is trained on January to September sales and tested on October to December. A random split would let it learn from December while predicting November.

REMEMBER

If the world changes over time, random splits flatter your model.

Three ways to cut one year of sales data



! Make the test window look like the period the model will serve, including its seasonality and promotions.

SECTION 2 · SPLITTING AND VALIDATION

Group splits

DEFINITION

A group split keeps every record belonging to one entity, such as a patient, customer or device, in the same set, so the model cannot recognise individuals instead of learning patterns.

EXAMPLE

A medical image model sees several scans per patient. If one patient’s scans appear in both training and test, the model can succeed by recognising that patient.

REMEMBER

Split by the unit you will predict for in the future.

How k fold cross validation works

01

Divide

Split the data into k equal parts, called folds.

02

Hold one

Keep one fold aside for validation.

03

Train

Fit the model on the remaining folds.

04

Repeat

Rotate so every fold is used once for validation.

05

Summarise

Report the average and spread across folds.

Five or ten folds are common. The spread across folds is as informative as the average: it shows how stable the model is.

Five folds, five rounds, one picture

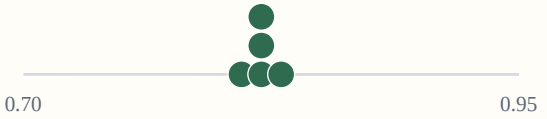
	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Score
Round 1	validate	train	train	train	train	0.81
Round 2	train	validate	train	train	train	0.84
Round 3	train	train	validate	train	train	0.79
Round 4	train	train	train	validate	train	0.86
Round 5	train	train	train	train	validate	0.82

Mean 0.824
spread (SD) ± 0.027

Every row is a fresh model trained from scratch on the four blue folds.

SAME MEAN, DIFFERENT STORY

Model A 0.82 ± 0.007



Tight cluster: dependable.

Model B 0.83 ± 0.08



Wide scatter: the 0.92 fold is luck.

! Report the mean and the spread. When spreads overlap this much, a 0.01 gap in means is not a win.

Cross validation variants

Variant	What it does	Use when
Stratified k fold	Keeps class balance in each fold	Classification, imbalanced data
Group k fold	Keeps each group in one fold	Repeated entities like patients
Time series split	Folds move forward in time	Forecasting and drifting data
Repeated k fold	Repeats with new shuffles	Small data, unstable estimates
Leave one out	Each row is its own fold	Very small datasets only
Nested	Inner loop tunes, outer loop scores	Tuning plus honest estimates

SECTION 2 · SPLITTING AND VALIDATION

Why nested cross validation matters

When you tune hyperparameters with cross validation and then report that same score, the score is optimistic because tuning chose the luckiest settings.

Inner loop

Searches for the best settings using only the outer training folds.

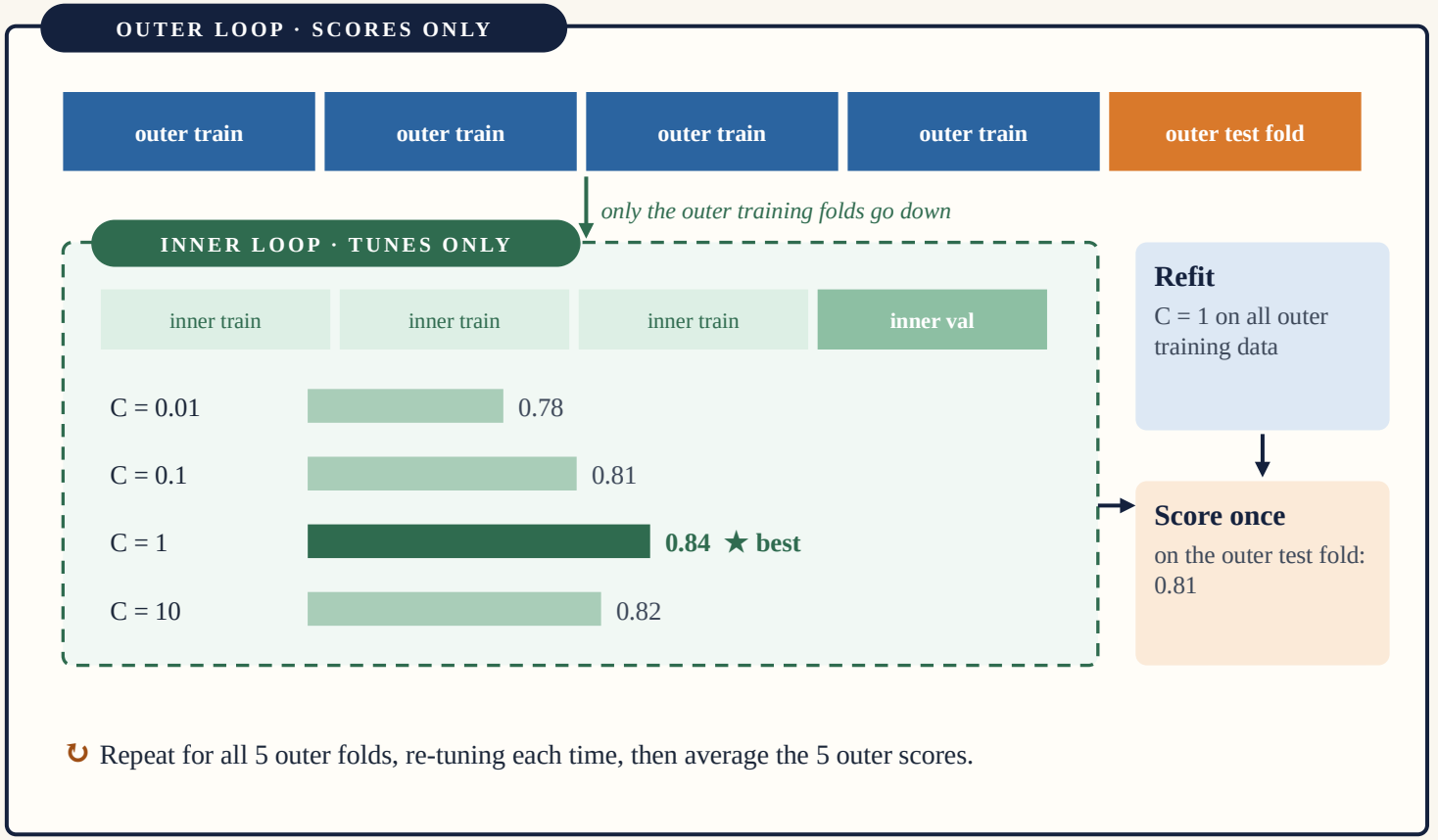
Outer loop

Scores the tuned model on folds the tuning never saw.

Result

An honest estimate of how the whole tuning procedure will perform.

Nested cross validation: a loop inside a loop



WHAT YOU WOULD REPORT

NON-NESTED

0.84

The best inner score: tuning picked the luckiest setting.

NESTED

0.81

Average of outer folds that tuning never touched.

REMEMBER

Nested CV scores the whole tuning procedure, not one fixed setting. The 0.03 gap is optimism.

Choosing split sizes

Dataset size	Typical approach	Reason
Hundreds of rows	Repeated cross validation	Every row is precious
Thousands	Cross validation plus small test set	Balance stability and honesty
Hundreds of thousands	Holdout of a few percent	Even small shares are large
Millions or more	Fixed holdout and fresh samples	Compute cost dominates



What matters is the number of examples of the rarest important case in each set, not the percentage.

Splitting pitfalls

PITFALL	BETTER PRACTICE
⚠ Random splits on time ordered data.	→ ✓ Train on the past and test on the future.
⚠ Duplicates spread across training and test.	→ ✓ Remove duplicates or split by group before evaluating.
⚠ Tuning repeatedly against the test set.	→ ✓ Tune on validation folds and open the test set once.
⚠ Preprocessing fitted on all data before splitting.	→ ✓ Fit scalers and encoders only on training folds.

Knowledge check

? A churn model uses three years of monthly customer snapshots. How should you split the data?

MODEL ANSWER

Use a time based split with groups in mind: train on earlier months, validate on a later period and test on the most recent months. Make sure a customer’s future snapshots never inform the model when scoring their past, and check results on customers new to the test period.

WHY IT MATTERS

It combines time and group thinking, which real data usually needs.

Section recap

KEY TAKEAWAYS

- Training, validation and test sets have distinct jobs
- Stratify for imbalanced classification
- Split by time when the future differs from the past
- Split by group to stop memorising entities
- Use cross validation for stable estimates
- Nest it when tuning to stay honest

TRY THIS

Review a past project's split. Identify any time, group or duplicate issue and estimate how it might have inflated results.

NEXT UP

Section 3: data leakage, the silent killer of evaluations.

SECTION 3

03

Data leakage

The most common reason a model looks brilliant in testing and fails in the real world.

IN THIS SECTION

- What leakage is
- Target leakage
- Train test contamination
- Temporal leakage
- Detecting leakage
- Prevention habits

“

If your results look too good to be true, assume leakage until you prove otherwise.

☆ Leakage gives the model access during evaluation to information it will not have when making real predictions.

SECTION 3 · DATA LEAKAGE

Target leakage

DEFINITION

Target leakage happens when a feature contains information created after, or because of, the outcome you are trying to predict.

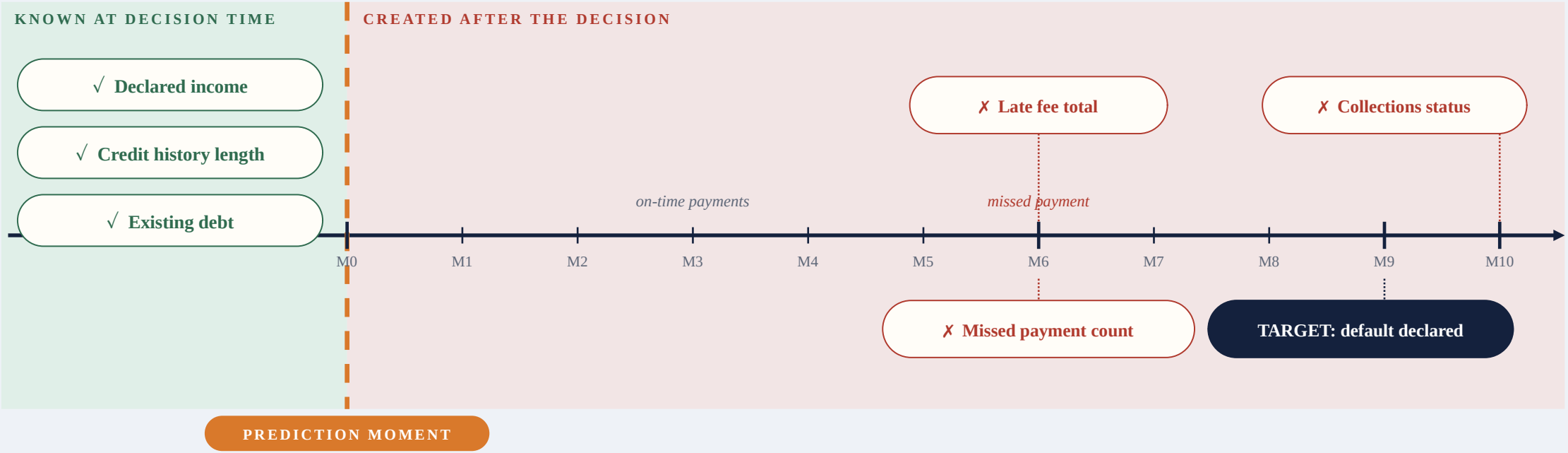
EXAMPLE

A model predicting loan default uses a field called collections status. Accounts only enter collections after default, so the feature reveals the answer.

REMEMBER

Ask of every feature: would I truly know this at prediction time?

Target leakage on the life of one loan



THE AVAILABILITY TEST

Would this value exist, with this exact content, at the prediction moment?

YES → KEEP

NO → DROP

Check the timestamp the value was written, not the date it describes.

WHY IT LEAKS

Accounts enter collections only after default. The feature is the label in disguise.

SECTION 3 · DATA LEAKAGE

Train test contamination

DEFINITION

Contamination happens when information from the test data influences training, for example through shared preprocessing, duplicates or feature selection done on all data.

EXAMPLE

A team scales features using the mean of the full dataset, then splits it. The test set has quietly shaped the training data.

REMEMBER

Split first. Fit every transformation only on training data.

SECTION 3 · DATA LEAKAGE

Temporal leakage

Temporal leakage uses information from the future relative to the moment a prediction would actually be made.

Future aggregates

A customer’s average spend computed over the whole year, including months after the prediction date.

Late corrections

Data fixed or backfilled later than it would have been available in real time.

Fix

Compute every feature as of a point in time, using only data available then.

SECTION 3 · DATA LEAKAGE

Warning signs of leakage

1 Near perfect scores

Performance far above published results or expert intuition.

2 One dominant feature

A single feature carries most of the predictive power.

3 Big offline online gap

Strong offline results collapse once the model goes live.

4 Strange feature names

Fields like status, outcome, resolution or final date.

5 Unstable across time

Performance drops sharply on newer periods.

6 Identifiers predict well

IDs or timestamps help the model, hinting at hidden structure.

SECTION 3 · DATA LEAKAGE

A leakage audit

1. **Map the timeline.** Mark when each feature is created relative to the prediction moment.
2. **Inspect top features.** Review the most important features with a domain expert.
3. **Drop and compare.** Remove suspicious features and see how much performance falls.
4. **Check duplicates.** Search for identical or near identical rows across sets.
5. **Recreate pipeline.** Confirm preprocessing is fitted inside each training fold only.
6. **Test forward.** Score on a later period that the model could not have seen.

Preprocessing inside the pipeline

no_leak_cv.py

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score, StratifiedKFold

pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(pipe, X, y, cv=cv, scoring="roc_auc")
print(scores.mean(), scores.std())
```

Inside each fold

The scaler is refit on training folds only, so validation data never shapes it.

Report spread

The standard deviation across folds shows stability, not just the average.

Same for selection

Feature selection and resampling also belong inside the pipeline.

Where you fit the scaler decides whether the test is honest

LEAKY ORDER

1. All rows together
2. **Fit the scaler on every row (test rows included)**
3. Split into training and test
4. Train and score: the test set has already shaped the features

TINY EXAMPLE

Training values
10, 20, 30

Test value
100

Mean used for scaling
40

≠

SAFE ORDER

1. Split into training and test first
2. Fit the scaler on training rows only
3. Transform the test set with the training mean and SD
4. Train and score: the test set stays unseen

TINY EXAMPLE

Training values
10, 20, 30

Test value
100

Mean used for scaling
20

```
make_pipeline(StandardScaler(), LogisticRegression())
```
← *cross_val_score refits the scaler inside every fold, so the safe order happens automatically*

FIT INSIDE EACH TRAINING FOLD, NEVER BEFORE THE SPLIT

Imputation · Scaling · Encoding · Feature selection · Resampling · Target encoding

SECTION 3 · DATA LEAKAGE

Leakage pitfalls

PITFALL	BETTER PRACTICE
⚠ Oversampling before splitting the data.	✓ Resample only inside training folds.
⚠ Selecting features using the whole dataset.	✓ Run feature selection within each fold.
⚠ Target encoding with the full target column.	✓ Use out of fold target encoding.
⚠ Trusting a feature because it is in the warehouse.	✓ Confirm when each value becomes available.

SECTION 3 · DATA LEAKAGE

Section recap

KEY TAKEAWAYS

- Leakage lets evaluation see information unavailable in real use
- Target leakage uses features caused by the outcome
- Contamination comes from shared preprocessing and duplicates
- Temporal leakage borrows from the future
- Suspiciously good results demand an audit
- Pipelines keep transformations inside each fold

TRY THIS

Take your current model's top ten features and write, for each, exactly when its value becomes available.

NEXT UP

Section 4: classification metrics.

SECTION 4

04

Classification metrics

Counting right and wrong answers in the ways that actually matter for decisions.

IN THIS SECTION

- The confusion matrix
- Accuracy and its limits
- Precision and recall
- Specificity and F1
- Balanced metrics
- Multiclass averaging

SECTION 4 · CLASSIFICATION

The confusion matrix

	Model predicts positive	Model predicts negative
Truly positive	True positive: correct alarm	False negative: missed case
Truly negative	False positive: false alarm	True negative: correct quiet

Almost every classification metric is a different way of summarising these four counts. Always look at the counts themselves, not just the summaries.

SECTION 4 · CLASSIFICATION

One confusion matrix, six metrics, all worked out

1,000 patients · 100 truly have the condition

	Predicted +	Predicted −
Truly +	TP **80** correct alarms	FN **20** missed cases
Truly −	FP **45** false alarms	TN **855** correct quiet

Colours in each formula match the cells they use.

Precision — 64%

TP ÷ (TP + FP)

80 ÷ 125

Of 125 alarms, 80 were real

Recall — 80%

TP ÷ (TP + FN)

80 ÷ 100

Of 100 real cases, 80 were caught

F1 score — 0.71

2 × P × R ÷ (P + R)

2 × 0.64 × 0.80 ÷ 1.44

Pulled toward the weaker of P and R

Accuracy — 93.5%

(TP + TN) ÷ all

935 ÷ 1,000

Flattered by 855 easy negatives

Specificity — 95%

TN ÷ (TN + FP)

855 ÷ 900

Healthy people correctly left alone

False positive rate — 5%

FP ÷ (FP + TN)

45 ÷ 900

Equal to 1 − specificity

TAKEAWAY Accuracy says 93.5%, yet 45 of 125 alarms are false. Always show the four counts beside the summaries.

SECTION 4 · CLASSIFICATION

Accuracy

DEFINITION

Accuracy is the share of all predictions that are correct, counting both positives and negatives together.

EXAMPLE

A model makes 1,000 predictions and gets 920 right, so its accuracy is 92 percent.

REMEMBER

Accuracy is misleading when one class is much more common than the other.

SECTION 4 · CLASSIFICATION

Precision

DEFINITION

Precision is the share of items the model flagged as positive that really are positive. It answers: when the model raises an alarm, how often is it right?

EXAMPLE

A fraud model flags 200 payments and 150 are genuinely fraudulent, so precision is 75 percent. The other 50 are genuine customers inconvenienced.

REMEMBER

High precision means few false alarms.

SECTION 4 · CLASSIFICATION

Recall, also called sensitivity

DEFINITION

Recall is the share of all truly positive items that the model successfully found. It answers: of everything we needed to catch, how much did we catch?

EXAMPLE

There were 300 fraudulent payments and the model caught 150, so recall is 50 percent. Half of the fraud slipped through.

REMEMBER

High recall means few missed cases.

SECTION 4 · CLASSIFICATION

The precision and recall trade off

FAVOUR PRECISION

When false alarms are costly

- Spam filters hiding real email
- Automatic account suspensions
- Expensive manual investigations
- Recommending products to buy

FAVOUR RECALL

When misses are costly

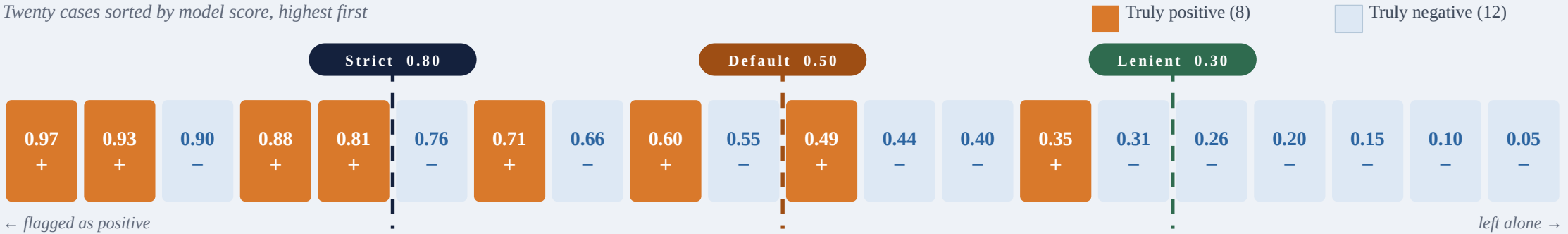
- Cancer screening
- Fraud and security threats
- Safety defects in manufacturing
- Child safety content detection

VERDICT You rarely get both for free. Raising the decision threshold usually lifts precision and lowers recall, so let error costs choose the balance.

SECTION 4 · CLASSIFICATION

Slide the threshold and watch precision and recall swap places

Twenty cases sorted by model score, highest first



Strict 0.80 — flags 5 · catches 4 of 8

Precision	4 ÷ 5	**80%**
Recall	4 ÷ 8	**50%**

Default 0.50 — flags 10 · catches 6 of 8

Precision	6 ÷ 10	**60%**
Recall	6 ÷ 8	**75%**

Lenient 0.30 — flags 15 · catches 8 of 8

Precision	8 ÷ 15	**53%**
Recall	8 ÷ 8	**100%**

TRADE OFF Moving the line right raises recall (50% → 100%) and lowers precision (80% → 53%). The cost of each error type decides where it stops.

SECTION 4 · CLASSIFICATION

More metrics from the confusion matrix

1 **Specificity**

Share of true negatives correctly left alone. The recall of the negative class.

2 **False positive rate**

Share of negatives wrongly flagged. One minus specificity.

3 **F1 score**

A single score that balances precision and recall, punishing either being low.

4 **F beta score**

Like F1 but weights recall more or less heavily than precision.

5 **Balanced accuracy**

Average of recall on each class, fair to rare classes.

6 **Matthews correlation**

Uses all four cells, robust when classes are imbalanced.

SECTION 4 · CLASSIFICATION

Understanding the F1 score

F1 combines precision and recall into one number that is high only when both are high. It behaves like a cautious average that is pulled down by the weaker of the two.

Strength

Useful when you care about the positive class and need one number for comparison.

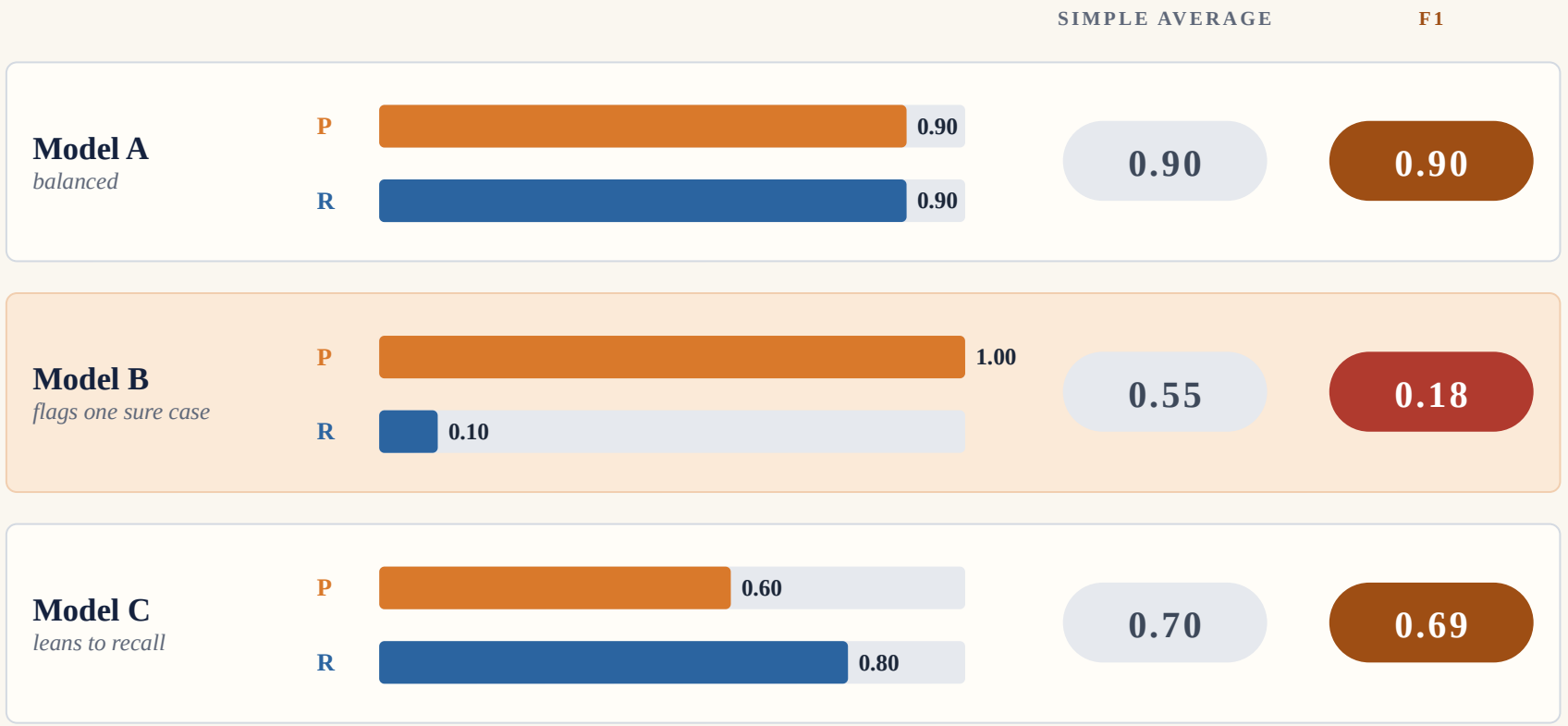
Weakness

Ignores true negatives and assumes precision and recall matter equally.

Tip

Report precision and recall beside F1 so readers can see which one is weak.

Why F1 uses a harmonic mean, not a simple average



Equal P and R: both averages agree. One collapses: only F1 notices.

THE FORMULA

F1 = 2PR ÷ (P + R)

Model B: 2 × 1.0 × 0.1 ÷ 1.1 = 0.18

NEED A TILT?

Use F beta

β = 2 counts recall twice as much as precision. β = 0.5 favours precision.

REMEMBER

Print P and R beside F1, so readers can see which one is weak.

SECTION 4 · CLASSIFICATION

Cohen’s kappa and agreement

Kappa measures how much a model agrees with the true labels beyond what chance agreement alone would produce.

Why useful

Corrects for the agreement you would get by guessing in proportion to class frequencies.

Also for humans

Commonly used to check how consistently human annotators label data.

Read it as

Near zero means chance level agreement. Near one means near perfect agreement.

SECTION 4 · CLASSIFICATION

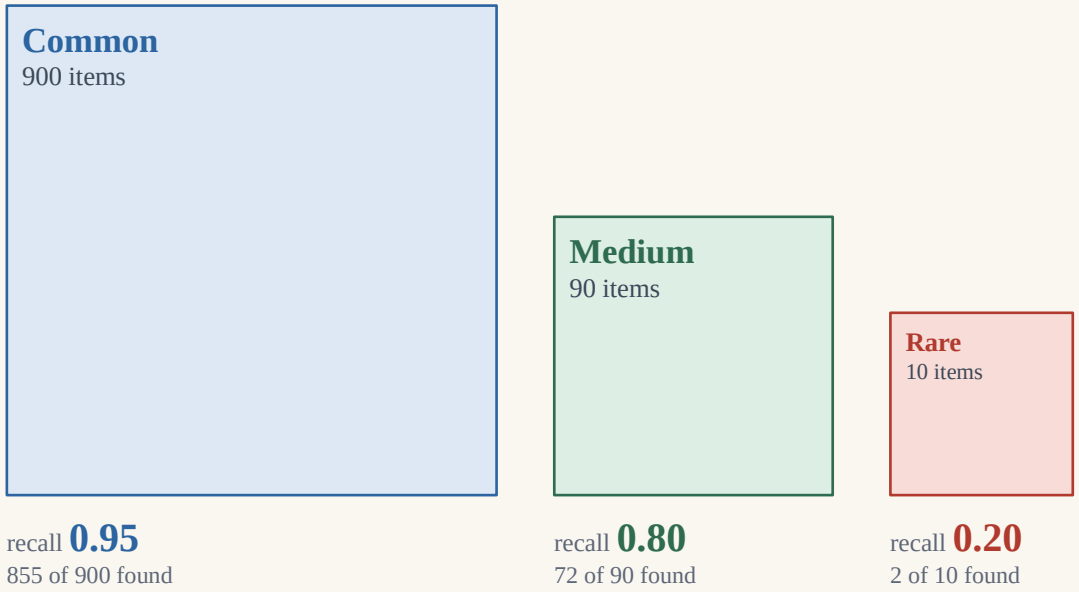
Averaging metrics across many classes

Averaging	How it works	Best when
Macro	Compute per class, then average equally	Every class matters equally
Weighted	Average per class weighted by class size	Reflecting overall population
Micro	Pool all decisions, then compute once	Overall decision quality matters
Per class	Report each class separately	Always, alongside any average

Macro averages expose weak performance on rare classes that micro and weighted averages can hide.

Macro, weighted and micro averages on one three-class example

BOX SIZE SHOWS HOW MANY ITEMS EACH CLASS HAS



recall **0.95**
855 of 900 found

recall **0.80**
72 of 90 found

recall **0.20**
2 of 10 found

REMEMBER

Macro is the only average here that notices the rare class failing.

Micro

Pool every decision, then compute once

(855 + 72 + 2) ÷ 1,000

0.929

Equals accuracy in single-label tasks

Weighted

Average per class, weighted by class size

0.90 × 0.95 + 0.09 × 0.80 + 0.01 × 0.20

0.929

Mirrors the population mix

Macro

Average per class, every class counts equally

(0.95 + 0.80 + 0.20) ÷ 3

0.65

Drags the rare-class failure into view

! Always print the per-class table beside any average: 0.95, 0.80 and 0.20 tell the real story.

SECTION 4 · CLASSIFICATION

Multilabel and top k accuracy

Some tasks allow several correct labels per item, or let the user choose among several suggestions, so plain accuracy is not enough.

Multilabel

Evaluate each label as its own yes or no task, then average, and report exact match rate.

Top k accuracy

Counts a prediction correct if the true label appears among the k most likely options.

Hamming loss

Share of individual label decisions that are wrong across all items.

Choosing a classification metric

Situation	Sensible primary metric
Balanced classes, equal error costs	Accuracy plus per class recall
False alarms are expensive	Precision at an agreed recall level
Missed cases are dangerous	Recall at an agreed precision level
Rare positive class	Precision recall curve area, F1, MCC
Many classes of equal importance	Macro averaged F1
Users choose from suggestions	Top k accuracy

SECTION 4 · CLASSIFICATION

Classification metric pitfalls

PITFALL	BETTER PRACTICE
Reporting accuracy on imbalanced data.	Report precision, recall and a baseline.
Quoting F1 without precision and recall.	Show both components alongside F1.
Using micro averages to hide weak classes.	Add macro averages and per class results.
Comparing metrics at different thresholds.	Fix the threshold or compare full curves.

SECTION 4 · CLASSIFICATION

Section recap

KEY TAKEAWAYS

- The confusion matrix underlies most classification metrics
- Precision limits false alarms, recall limits misses
- Error costs decide the right balance
- F1, balanced accuracy and MCC summarise imbalanced results
- Choose macro, micro or weighted averages deliberately
- Always report per class results

TRY THIS

Compute the confusion matrix, precision, recall, F1 and balanced accuracy for one of your models, then explain each number in one sentence.

NEXT UP

Section 5: thresholds, probabilities and calibration.

SECTION 5

05

Thresholds, probabilities and calibration

Evaluating the scores behind decisions, not just the decisions themselves.

IN THIS SECTION

- Scores and thresholds
- ROC curves and AUC
- Precision recall curves
- Choosing a threshold
- Log loss and Brier score
- Calibration

SECTION 5 · PROBABILITIES

Scores versus decisions

Most classifiers output a score or probability. A threshold turns that score into a yes or no decision, and different thresholds give different metrics.

Ranking quality

How well the scores order positives above negatives, regardless of threshold.

Decision quality

How good the yes or no choices are at one chosen threshold.

Probability quality

Whether a score of 70 percent really means 7 in 10 such cases are positive.

SECTION 5 · PROBABILITIES

The ROC curve

DEFINITION

The ROC curve plots the true positive rate against the false positive rate as the decision threshold moves from strict to lenient, showing every trade off the model offers.

EXAMPLE

At a strict threshold a screening model catches 40 percent of cases with 2 percent false alarms. At a lenient one it catches 90 percent with 20 percent false alarms.

REMEMBER

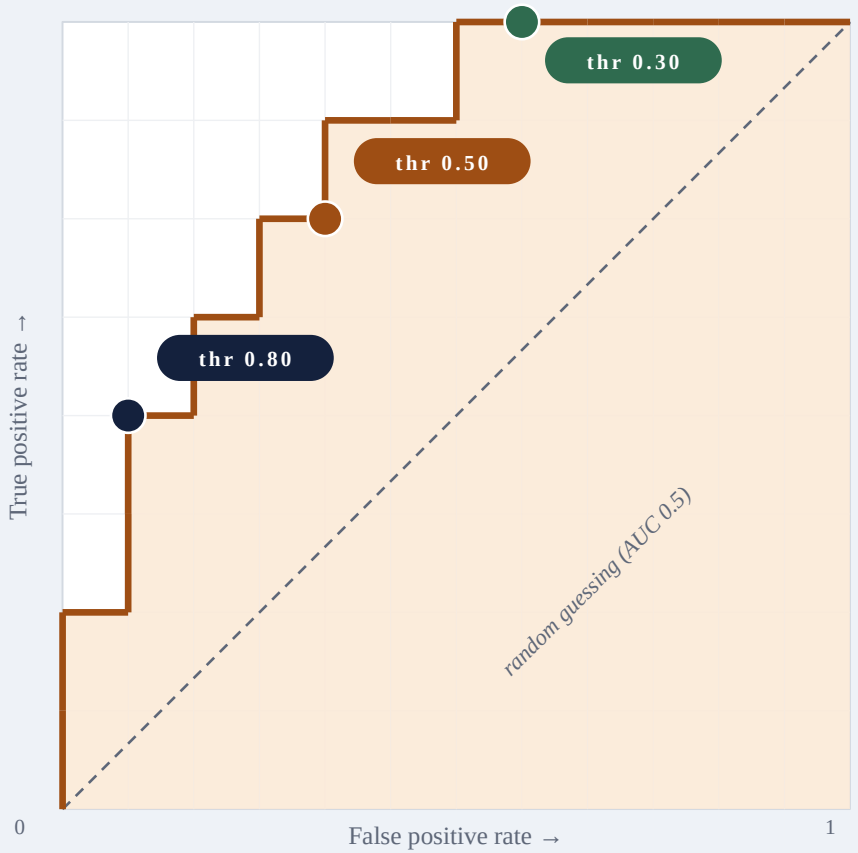
A curve that hugs the top left corner is better. The diagonal means random guessing.

Building an ROC curve by walking down the ranking

WALK DOWN THE RANKING

0.97 + ↑ up ⅛	**0.49** + ↑ up ⅛
0.93 + ↑ up ⅛	**0.44** − → right 1/12
0.90 − → right 1/12	**0.40** − → right 1/12
0.88 + ↑ up ⅛	**0.35** + ↑ up ⅛
0.81 + ↑ up ⅛	**0.31** − → right 1/12
0.76 − → right 1/12	**0.26** − → right 1/12
0.71 + ↑ up ⅛	**0.20** − → right 1/12
0.66 − → right 1/12	**0.15** − → right 1/12
0.60 + ↑ up ⅛	**0.10** − → right 1/12
0.55 − → right 1/12	**0.05** − → right 1/12

Same 20 cases as the threshold slide: 8 positives, 12 negatives.



SHADED AREA = AUC

0.82

79 of 96 positive–negative pairs are ranked in the right order.

READING THE CORNERS

Bottom left: flag nobody.
Top right: flag everybody.
Top left corner: perfect ranking.

REMEMBER

The curve shows every threshold at once. AUC never picks one for you.

SECTION 5 · PROBABILITIES

ROC AUC

DEFINITION

The area under the ROC curve summarises ranking quality as one number: the chance that a randomly chosen positive gets a higher score than a randomly chosen negative.

EXAMPLE

An AUC of 0.85 means that in 85 percent of random positive and negative pairs, the positive case is ranked higher.

REMEMBER

0.5 is random ranking, 1.0 is perfect ranking. AUC says nothing about calibration.

SECTION 5 · PROBABILITIES

ROC curve versus precision recall curve

ROC

True positive rate against false positive rate

- Insensitive to class balance
- Can look optimistic with rare positives
- Good for comparing ranking ability
- Summarised by ROC AUC

PRECISION RECALL

Precision against recall

- Focuses on the positive class
- Reveals false alarm burden with rare positives
- Good for imbalanced, alarm style tasks
- Summarised by average precision

VERDICT When positives are rare and false alarms matter, prefer the precision recall curve. Report both when in doubt.

SECTION 5 · PROBABILITIES

Choosing a decision threshold

1. **Never assume 0.5.** The default threshold is rarely the best one for your costs.
2. **Use validation data.** Choose thresholds on validation data, never the test set.
3. **Apply constraints.** For example, the highest recall with precision of at least 80 percent.
4. **Use costs.** Pick the threshold that minimises total expected cost of errors.
5. **Consider capacity.** Match the number flagged to how many cases staff can review.
6. **Recheck over time.** Thresholds drift as data and base rates change.

SECTION 5 · PROBABILITIES

Cost sensitive evaluation

Assign a realistic cost to each kind of error and benefit to each correct decision, then evaluate models by the total cost or value they would create.

Cost matrix

Write down the cost of a false positive and of a false negative with the business.

Expected value

Multiply counts by costs and add them up for each candidate threshold.

Communicates well

Money or hours saved is easier for stakeholders to understand than AUC.

Two fraud models, two receipts: let the cost matrix choose

AGREED COST MATRIX

	Flag	Pass
Fraud	**£0** fraud stopped	**£500** missed fraud
Genuine	**£20** false alarm call	**£0** smooth payment

A miss costs 25 times a false alarm.

Scenario: 10,000 transactions, 100 of them fraud.

NOTICE

Model B raises eight times as many alarms, yet costs £5,500 less.

Model A · cautious

accuracy 99.1% · precision 55% · recall 60%

Caught	60 frauds	£0
Missed	40 × £500	£20,000
Alarms	50 × £20	£1,000
TOTAL COST		**£21,000**

per 10,000 payments

Model B · eager

accuracy 95.9% · precision 18% · recall 85%

Caught	85 frauds	£0
Missed	15 × £500	£7,500
Alarms	400 × £20	£8,000
TOTAL COST		**£15,500**

per 10,000 payments

WINNER

VERDICT Accuracy and precision pick Model A. The business ledger picks Model B. Report the ledger.

SECTION 5 · PROBABILITIES

Scoring probabilities directly

1 Log loss

Heavily punishes confident wrong predictions. Lower is better.

2 Brier score

Average squared gap between predicted probability and outcome. Lower is better.

3 Why use them

They reward honest probabilities, not just correct rankings.

4 Proper scores

Both are proper scoring rules: the best score comes from true probabilities.

5 Compare to baseline

Always compare with predicting the base rate for everyone.

6 Sensitive to rare cases

A few very confident mistakes can dominate log loss.

SECTION 5 · PROBABILITIES

Calibration

DEFINITION

A model is well calibrated when its predicted probabilities match observed frequencies: among cases given 30 percent, about 30 percent turn out positive.

EXAMPLE

A weather model says 70 percent chance of rain on 100 days. If it rains on about 70 of them, the model is well calibrated.

REMEMBER

Good ranking does not guarantee good calibration. Check both.

SECTION 5 · PROBABILITIES

Reading a reliability diagram

01

Bin

Group predictions into bins by predicted probability.

02

Average

Compute the mean predicted probability in each bin.

03

Observe

Compute the actual share of positives in each bin.

04

Plot

Draw observed against predicted for every bin.

05

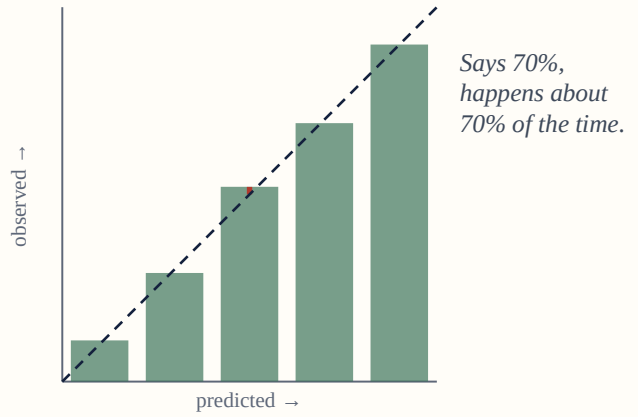
Judge

Points on the diagonal mean good calibration.

Points below the diagonal mean overconfidence. Points above it mean underconfidence. Expected calibration error summarises the average gap.

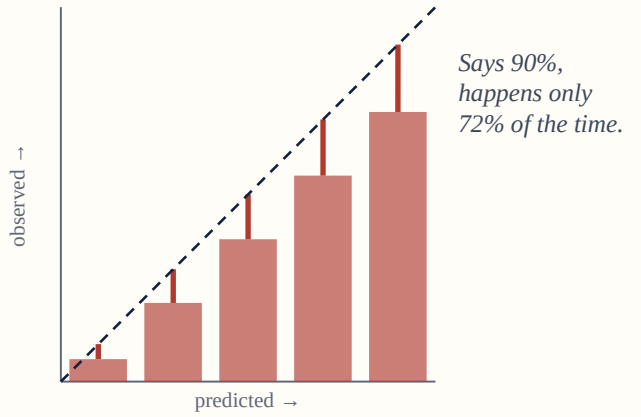
Three reliability diagrams and how to read the gaps

Well calibrated



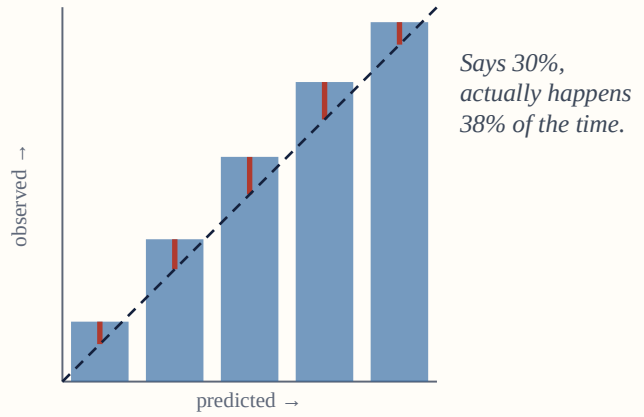
Says 70%, happens about 70% of the time.

Overconfident



Says 90%, happens only 72% of the time.

Underconfident



Says 30%, actually happens 38% of the time.

EXPECTED CALIBRATION ERROR, WORKED FOR THE OVERCONFIDENT MODEL

Predicted (bin mean)	.10	.30	.50	.70	.90
Observed share	.06	.21	.38	.55	.72
Gap	**.04**	**.09**	**.12**	**.15**	**.18**

ECE

0.116

mean gap, equal bins

FIX

Recalibrate on held-out data (Platt, isotonic or temperature scaling), then re-check this diagram.

SECTION 5 · PROBABILITIES

Methods to improve calibration

Method	Idea	Best for
Platt scaling	Fits a simple curve mapping scores to probabilities	Small datasets, smooth errors
Isotonic regression	Fits a flexible stepwise mapping	Larger datasets
Temperature scaling	Softens or sharpens neural network outputs	Deep learning classifiers
Recalibrate regularly	Refit on recent data as base rates shift	Drifting environments



Always fit calibration on data not used to train the model, then confirm the effect on a separate test set.

SECTION 5 · PROBABILITIES

Threshold and probability pitfalls

PITFALL	BETTER PRACTICE
⚠ Using 0.5 because it is the default.	✓ Choose thresholds from costs or constraints.
⚠ Treating scores as true probabilities.	✓ Check calibration before using scores as risks.
⚠ Celebrating high AUC on rare positives.	✓ Check the precision recall curve as well.
⚠ Tuning thresholds on the test set.	✓ Tune on validation and confirm on test once.

SECTION 5 · PROBABILITIES

Section recap

KEY TAKEAWAYS

- Separate ranking, decision and probability quality
- ROC AUC measures ranking across thresholds
- Precision recall curves suit rare positives
- Choose thresholds with costs and capacity
- Log loss and Brier score reward honest probabilities
- Check and fix calibration with reliability diagrams

TRY THIS

Plot ROC, precision recall and reliability diagrams for your model, then choose a threshold using a simple cost matrix.

NEXT UP

Section 6: regression metrics.

SECTION 6

06

Regression metrics

Measuring how far numeric predictions land from reality, and which errors matter most.

IN THIS SECTION

- Absolute and squared errors
- Percentage errors
- Explained variance
- Quantile and interval scores
- Residual analysis
- Choosing a metric

SECTION 6 · REGRESSION

Mean absolute error

DEFINITION

Mean absolute error is the average size of the prediction errors, ignoring direction. It is expressed in the same units as the target.

EXAMPLE

A house price model is off by 10,000, 5,000 and 30,000 pounds on three homes. Its mean absolute error is 15,000 pounds.

REMEMBER

Easy to explain. Every pound of error counts the same.

SECTION 6 · REGRESSION

Root mean squared error

DEFINITION

Root mean squared error squares each error, averages them and takes the square root, so large errors count much more than small ones. It stays in the target’s units.

EXAMPLE

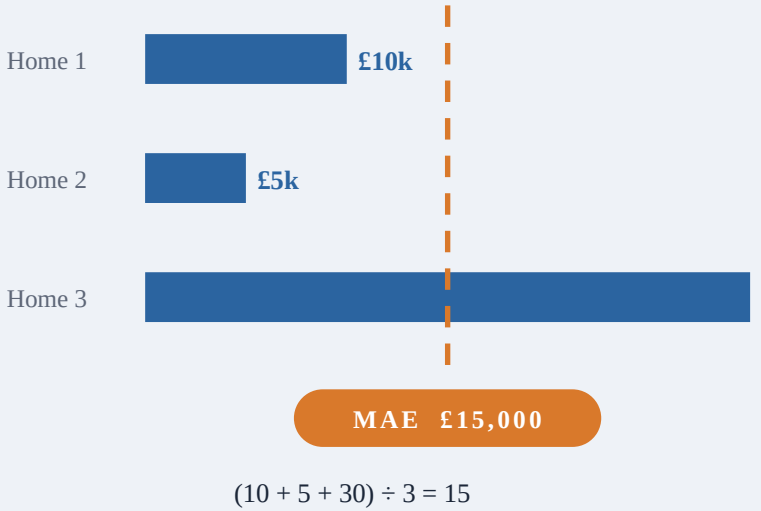
For the same three homes, the 30,000 pound miss dominates, making this error larger than the mean absolute error.

REMEMBER

Use it when large mistakes are disproportionately harmful.

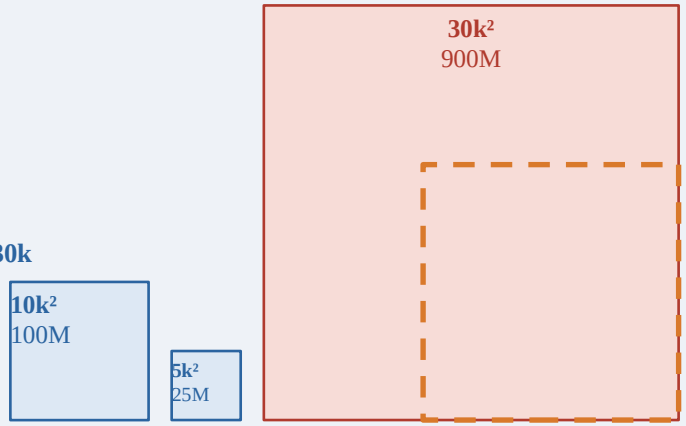
MAE averages lengths; RMSE averages areas

ABSOLUTE ERRORS ARE LENGTHS



(10 + 5 + 30) ÷ 3 = 15

SQUARED ERRORS ARE AREAS



average area 341.7M
side = √341.7M ≈ 18.5k

RMSE £18,484

The dashed square has the average area. Its side is the RMSE.

WHAT IF HOME 3 MISSED BY £90K?

	Now	If 90k
MAE	**15.0k**	**35.0k**
RMSE	**18.5k**	**52.4k**

MAE grows ×2.3
RMSE grows ×2.8
Squaring amplifies the single worst miss.

READ THE GAP RMSE close to MAE: errors are similar sizes. RMSE far above MAE: a few big misses dominate, so go and find them.

SECTION 6 · REGRESSION

Absolute versus squared error

MAE

Mean absolute error

- Treats all errors in proportion
- Robust to a few outliers
- Matches median focused predictions
- Very easy to communicate

RMSE

Root mean squared error

- Penalises big errors heavily
- Sensitive to outliers
- Matches mean focused predictions
- Standard in many benchmarks

VERDICT Report both. A large gap between them signals a few big misses worth investigating.

SECTION 6 · REGRESSION

Percentage errors and their traps

Mean absolute percentage error expresses error relative to the true value, which is intuitive but breaks down in important cases.

Near zero values

Tiny actual values create huge or infinite percentages.

Asymmetry

It punishes over prediction and under prediction differently.

Alternatives

Weighted percentage error, symmetric versions or scaled errors are often safer.

SECTION 6 · REGRESSION

R squared

DEFINITION

R squared describes how much of the variation in the target the model explains, compared with simply predicting the average every time.

EXAMPLE

An R squared of 0.80 means the model removes about 80 percent of the error you would get by always predicting the mean.

REMEMBER

It can be negative on new data and says nothing about error size in real units.

SECTION 6 · REGRESSION

Other useful regression measures

1 Median absolute error

Typical error, unaffected by a few extreme misses.

2 Max error

The worst single miss, vital for safety limits.

3 Mean error, or bias

Shows if predictions are systematically too high or too low.

4 Log based errors

Focus on relative errors for targets spanning many magnitudes.

5 Error within tolerance

Share of predictions within an acceptable range.

6 Business cost

Translate errors into money, stock or time lost.

SECTION 6 · REGRESSION

Evaluating quantiles and intervals

Many decisions need a range rather than a single number. Quantile and interval predictions need their own evaluation methods.

Pinball loss

Scores quantile predictions, penalising misses on the wrong side more heavily.

Coverage

Share of true values falling inside the predicted interval, compared with the promised level.

Sharpness

Narrower intervals are more useful, as long as coverage is honest.

SECTION 6 · REGRESSION

Residual analysis

1. **Plot residuals against predictions.** Look for patterns such as funnels or curves.
2. **Plot against key features.** Find regions where the model is systematically wrong.
3. **Check over time.** Look for drift or seasonal patterns in errors.
4. **Inspect the largest errors.** Read the actual records behind the biggest misses.
5. **Check distribution.** See whether errors are skewed or have heavy tails.
6. **Act.** Add features, transform the target or segment the model.

SECTION 6 · REGRESSION

Regression metric pitfalls

PITFALL	BETTER PRACTICE
⚠ Using percentage error when values can be near zero.	→ ✓ Use scaled or weighted alternatives.
⚠ Reporting R squared alone.	→ ✓ Add error metrics in the target’s own units.
⚠ Ignoring systematic bias.	→ ✓ Report mean error and residual plots.
⚠ Averaging over very different segments.	→ ✓ Report errors per segment and size.

SECTION 6 · REGRESSION

Section recap

KEY TAKEAWAYS

- MAE is robust and easy to explain
- RMSE emphasises large errors
- Percentage errors break near zero
- R squared compares against predicting the mean
- Evaluate intervals with coverage and sharpness
- Residual plots reveal where models fail

TRY THIS

Compute MAE, RMSE, median error and bias for your regression model, then plot residuals against the top three features.

NEXT UP

Section 7: evaluating models on imbalanced data.

SECTION 7

07

Imbalanced data

When the cases that matter most are also the rarest, evaluation needs special care.

IN THIS SECTION

- Why imbalance distorts metrics
- Metrics that cope
- Resampling and evaluation
- Base rates and prevalence
- Capacity based evaluation

SECTION 7 · IMBALANCED DATA

Why imbalance distorts evaluation

When one class is 99 percent of the data, a model can look excellent while never finding the cases you care about.

Accuracy inflates

Predicting the majority class gives very high accuracy for free.

ROC looks rosy

Huge numbers of negatives make false positive rates look small.

Few positives

Metrics on rare classes become noisy and unstable.

Metrics that cope with imbalance

Metric	Why it helps
Precision and recall	Focus directly on the rare positive class
Average precision	Summarises the precision recall curve
F1 or F beta	Balances precision and recall in one number
Balanced accuracy	Gives each class equal weight
Matthews correlation	Uses all four confusion counts fairly
Recall at fixed precision	Matches real operating constraints

SECTION 7 · IMBALANCED DATA

Resampling and honest evaluation

Oversampling, undersampling and synthetic methods such as SMOTE can help training, but they must never change the evaluation data.

Resample training only

Apply resampling inside training folds after splitting.

Keep test realistic

The test set must reflect the true class balance seen in production.

Recalibrate

Resampling distorts probabilities, so recalibrate before using them as risks.

SECTION 7 · IMBALANCED DATA

Prevalence and base rates

DEFINITION

Prevalence is how common the positive class is. Precision depends strongly on it, so the same model can show very different precision in different populations.

EXAMPLE

A test with fixed recall and false positive rate gives high precision in a specialist clinic where the disease is common, but low precision in general screening.

REMEMBER

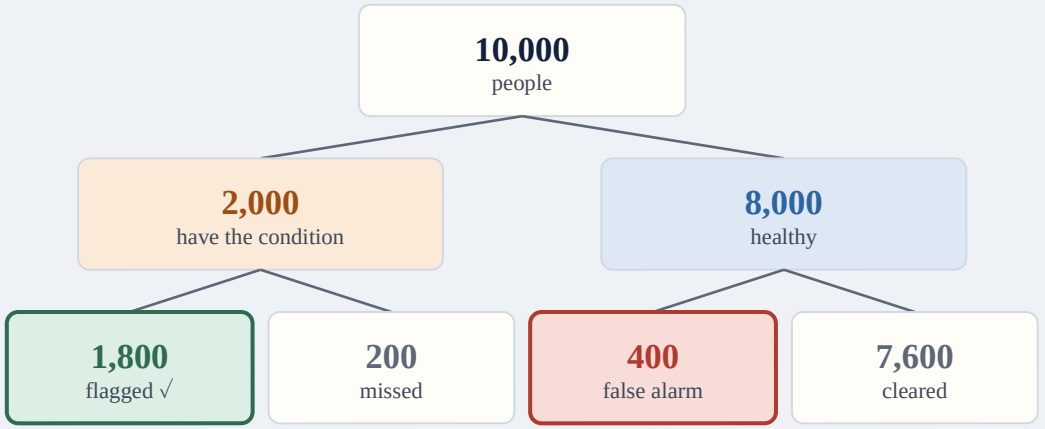
Always state the prevalence when you report precision.

Same model, two populations, very different precision

SAME MODEL IN BOTH: recall 90% · false positive rate 5%

Specialist clinic

20% prevalence

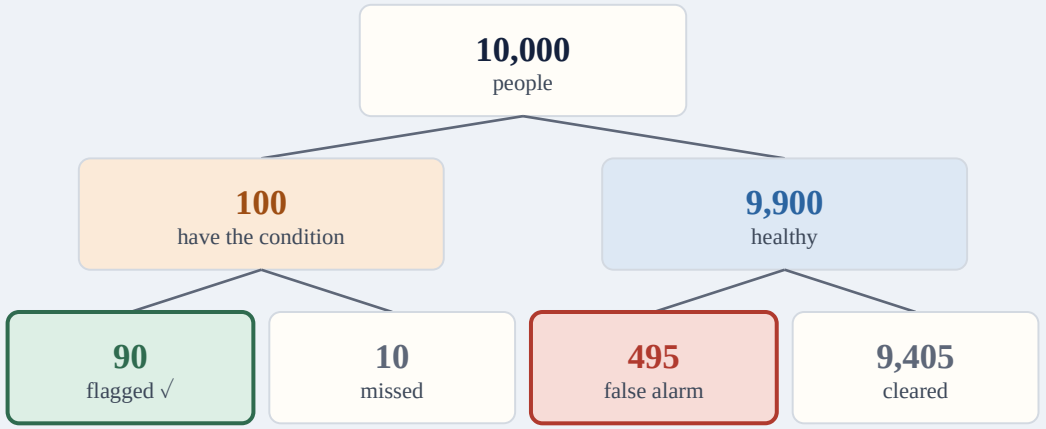


Precision 1,800 ÷ (1,800 + 400)
of flagged people truly have it

82%

General screening

1% prevalence



Precision 90 ÷ (90 + 495)
of flagged people truly have it

15%

! Precision travels badly between populations. Re-estimate it on data with the deployment prevalence.

SECTION 7 · IMBALANCED DATA

Evaluating under limited review capacity

Often staff can only review a fixed number of alerts per day. Evaluate the model the way it will be used: by what it finds within that budget.

Precision at k

Of the top k cases flagged each day, how many are real?

Recall at k

What share of all true cases are found within the top k?

Lift

How many times better than random selection the top flagged cases are.

SECTION 7 · IMBALANCED DATA

Imbalance pitfalls

PITFALL	BETTER PRACTICE
⚠ Evaluating on a rebalanced test set.	✓ Keep the test set at realistic prevalence.
⚠ Quoting precision without prevalence.	✓ State the base rate with every precision figure.
⚠ Too few positives to measure reliably.	✓ Use confidence intervals and gather more positives.
⚠ Using probabilities after resampling.	✓ Recalibrate before treating scores as risks.

SECTION 7 · IMBALANCED DATA

Knowledge check

? Investigators can review 100 alerts a day out of 50,000 transactions. Which metric best reflects the model's usefulness?

MODEL ANSWER

Precision at 100 and recall at 100: how many of the top 100 daily alerts are real fraud, and what share of all fraud they capture. These match the actual workflow far better than accuracy or overall AUC.

WHY IT MATTERS

Evaluating at the real operating point links metrics to work done.

Section recap

KEY TAKEAWAYS

- Imbalance makes accuracy and ROC look misleadingly good
- Use precision, recall, average precision and MCC
- Resample only training data, never test data
- Precision depends on prevalence
- Evaluate at the real review capacity

TRY THIS

For an imbalanced problem, report precision and recall at the top 1, 5 and 10 percent of scores and compare with random selection.

NEXT UP

Section 8: ranking and recommendation metrics.

SECTION 8

08

Ranking and recommendation

When the order of results matters as much as the results themselves.

IN THIS SECTION

- Why ranking is different
- Precision and recall at k
- MRR and MAP
- NDCG
- Beyond accuracy
- Offline versus online gaps

SECTION 8 · RANKING

Why ranking needs its own metrics

Users look at the top few results and rarely scroll. A good ranker puts the most relevant items first, so metrics must reward position.

Position matters

A relevant item in first place is worth more than the same item in twentieth.

Cut offs

Metrics are usually computed at k, the number of results a user actually sees.

Relevance levels

Items can be graded, such as perfect, good, fair or irrelevant.

SECTION 8 · RANKING

Core ranking metrics

1 Precision at k

Share of the top k results that are relevant.

2 Recall at k

Share of all relevant items that appear in the top k.

3 Hit rate at k

Share of users with at least one relevant item in the top k.

4 Mean reciprocal rank

Rewards placing the first relevant result as high as possible.

5 Mean average precision

Rewards placing all relevant results high, averaged over queries.

6 NDCG

Rewards graded relevance with a discount for lower positions.

SECTION 8 · RANKING

Normalised discounted cumulative gain

DEFINITION

NDCG adds up the relevance of results, giving less credit the further down the list each result appears, then divides by the best possible score so results fall between zero and one.

EXAMPLE

Two search engines return the same five relevant pages, but one puts the best page first. That engine earns a higher NDCG.

REMEMBER

NDCG is the standard choice when relevance comes in grades.

NDCG, calculated position by position

	Position 1	Position 2	Position 3	Position 4	Position 5	DCG	NDCG
Discount $1 \div \log_2(\text{position} + 1)$	1.00	0.63	0.50	0.43	0.39	*credit fades with position*	
Ideal order *best possible*	**grade 3** 3 × 1.00 = 3.00	**grade 3** 3 × 0.63 = 1.89	**grade 2** 2 × 0.50 = 1.00	**grade 1** 1 × 0.43 = 0.43	**grade 0** 0 × 0.39 = 0.00	**6.32**	**1.00**
Engine A *strong pages first*	**grade 3** 3 × 1.00 = 3.00	**grade 2** 2 × 0.63 = 1.26	**grade 3** 3 × 0.50 = 1.50	**grade 0** 0 × 0.43 = 0.00	**grade 1** 1 × 0.39 = 0.39	**6.15**	**0.97**
Engine B *same pages, buried*	**grade 0** 0 × 1.00 = 0.00	**grade 1** 1 × 0.63 = 0.63	**grade 3** 3 × 0.50 = 1.50	**grade 2** 2 × 0.43 = 0.86	**grade 3** 3 × 0.39 = 1.16	**4.15**	**0.66**

NDCG = DCG ÷ ideal DCG. Engine A: 6.15 ÷ 6.32 = 0.97. Engine B: 4.15 ÷ 6.32 = 0.66.

! Gain here is the grade itself. Many libraries use 2^grade − 1, which rewards top grades more, so say which one you used.

SECTION 8 · RANKING

Beyond accuracy for recommenders

1 Coverage
Share of the catalogue that ever gets recommended.

2 Diversity
How different the items in one list are from each other.

3 Novelty
How unfamiliar or less popular the recommendations are.

4 Serendipity
Relevant surprises users would not have found themselves.

5 Popularity bias
Whether the system keeps pushing already popular items.

6 Freshness
How quickly new items start being recommended.

SECTION 8 · RANKING

Why offline ranking results often disagree with online

⚠ Historical logs only contain feedback on items users were shown, so offline evaluation is biased towards the old system's choices.

Exposure bias

Items never shown have no clicks, even if they were great.

Position bias

Users click top results partly because they are on top.

Remedies

Counterfactual evaluation, interleaving experiments and A/B tests.

SECTION 8 · RANKING

Interleaving

DEFINITION

Interleaving merges results from two rankers into one list for each user and measures which ranker’s items get more clicks, giving a sensitive comparison with little traffic.

EXAMPLE

A search team blends results from the old and new rankers. Users click new ranker items more often, so it wins with far fewer users than an A/B test would need.

REMEMBER

Interleaving detects preference quickly but does not measure overall business impact.

SECTION 8 · RANKING

Ranking evaluation pitfalls

PITFALL	BETTER PRACTICE
⚠ Evaluating at k far larger than users see.	✓ Choose k from real interface behaviour.
⚠ Treating unclicked items as irrelevant.	✓ Account for items users never saw.
⚠ Optimising only clicks.	✓ Add diversity, coverage and long term engagement.
⚠ Random splits of interaction logs.	✓ Split by time to mimic future recommendations.

SECTION 8 · RANKING

Knowledge check

? A new recommender lifts precision at 10 offline but recommends only the 200 most popular products. Is it ready?

MODEL ANSWER

Probably not. It may be exploiting popularity bias, shrinking catalogue coverage and hurting discovery for niche products and new sellers. Check coverage, diversity and novelty, then confirm with an online test measuring long term engagement and revenue.

WHY IT MATTERS

Accuracy alone can reward a recommender that narrows choice.

Section recap

KEY TAKEAWAYS

- Ranking metrics reward correct order and position
- Use precision, recall and hit rate at a realistic k
- MRR, MAP and NDCG capture position effects
- Measure coverage, diversity and novelty too
- Offline logs carry exposure and position bias
- Interleaving and A/B tests confirm offline gains

TRY THIS

Compute precision at 10, NDCG at 10 and catalogue coverage for a recommender, then explain any trade offs you see.

NEXT UP

Section 9: evaluating forecasts.

SECTION 9

09

Forecasting evaluation

Judging predictions of the future without accidentally peeking at it.

IN THIS SECTION

- Why forecasting is special
- Backtesting
- Naive baselines
- Scaled errors
- Forecast horizons
- Probabilistic forecasts

SECTION 9 · FORECASTING

What makes forecasting different



Time only flows one way. Every evaluation must respect that the future is unknown when the forecast is made.

Order matters

Observations depend on earlier ones, so random shuffling breaks the problem.

Conditions change

Trends, seasons and shocks shift over time.

Many series

Forecasts often cover thousands of products or locations at once.

Rolling origin backtesting

01

Origin

Choose a cut off date and train only on data before it.

02

Forecast

Predict the next period, such as the next four weeks.

03

Score

Compare the forecast with what actually happened.

04

Roll

Move the cut off forward and repeat.

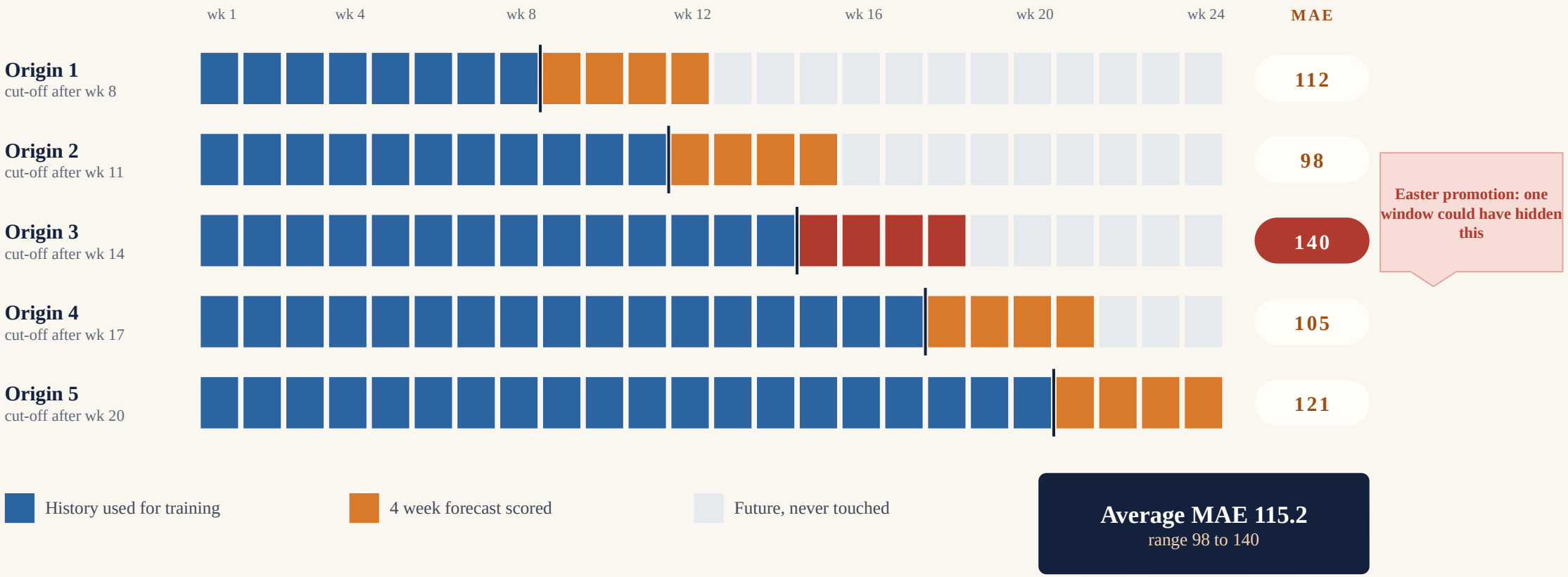
05

Aggregate

Average errors across all origins and horizons.

Backtesting over many origins shows how the model behaves across seasons, promotions and unusual periods, not just one lucky window.

Rolling origin backtesting across 24 weeks



! This is an expanding window. A sliding window drops the oldest weeks so every training history has the same length.

SECTION 9 · FORECASTING

Naive baselines are hard to beat

Simple forecasts are surprisingly strong. Every forecasting model must be compared with them before any claim of skill.

Last value

Tomorrow equals today. Strong for slowly changing series.

Seasonal naive

Next Monday equals last Monday. Strong for weekly and yearly cycles.

Moving average

The average of recent periods. Smooths noise.

Scaled errors

DEFINITION

A scaled error divides a model’s error by the error of a naive baseline on the same series, so values below one mean the model beats the baseline.

EXAMPLE

A product’s forecast scores 0.7, meaning its errors are 30 percent smaller than simply repeating last season’s sales.

REMEMBER

Scaled errors let you compare accuracy fairly across series of very different sizes.

A scaled error of 0.7, worked out week by week

	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Mean
Actual sales	120	135	128	150	142	160	
Naive: repeat last week	110	120	135	128	150	142	
Naive miss	**10**	**15**	**7**	**22**	**8**	**18**	**13.3**
Model forecast	128	125	134	136	136	148	
Model miss	**8**	**10**	**6**	**14**	**6**	**12**	**9.3**

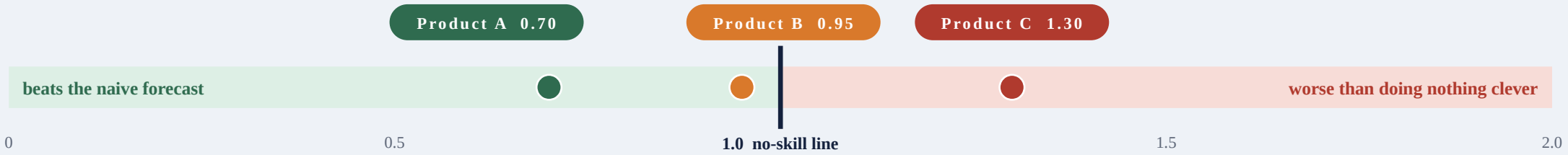
SCALED ERROR

9.3 ÷ 13.3

= 0.70

Errors 30% smaller than repeating last week.

ONE SCALE FOR EVERY PRODUCT



Product B is barely better than naive. Product C should fall back to the naive forecast.

Common forecasting metrics

Metric	What it captures	Watch for
MAE	Average size of misses in units	Hard to compare across series
RMSE	Emphasises big misses	Sensitive to outliers
MAPE	Error as a share of actuals	Breaks with zeros
Weighted MAPE	Total error over total volume	Dominated by big series
MASE	Error scaled by naive baseline	Needs a sensible baseline
Pinball loss	Quality of quantile forecasts	One score per quantile

SECTION 9 · FORECASTING

Forecast horizons

Accuracy usually falls the further ahead you forecast. Evaluate each horizon separately because different decisions depend on different horizons.

Short horizon

Next day staffing needs high accuracy in the very near term.

Medium horizon

Ordering stock weeks ahead depends on the four to eight week view.

Long horizon

Capacity planning months ahead tolerates more error but needs the right trend.

SECTION 9 · FORECASTING

Evaluating probabilistic forecasts

A forecast range, such as 80 percent sure demand will be between 900 and 1,200 units, supports better decisions than a single number.

Coverage

Do 80 percent intervals contain the truth about 80 percent of the time?

Sharpness

Are intervals as narrow as possible while staying honest?

Overall scores

Pinball loss and continuous ranked probability score combine both.

SECTION 9 · FORECASTING

Forecasting pitfalls

PITFALL	BETTER PRACTICE
⚠ Random cross validation on time series.	✓ Use rolling origin backtesting.
⚠ No comparison with naive forecasts.	✓ Always report scaled errors against baselines.
⚠ One average over all horizons.	✓ Report accuracy per horizon that matters.
⚠ Using revised data that was not available then.	✓ Evaluate with data as it was known at each origin.

SECTION 9 · FORECASTING

Section recap

KEY TAKEAWAYS

- Respect time order in every evaluation
- Backtest across many rolling origins
- Beat naive and seasonal naive baselines first
- Use scaled errors to compare across series
- Evaluate each decision horizon separately
- Check coverage and sharpness of forecast ranges

TRY THIS

Backtest your forecast over at least six origins and compare it with a seasonal naive baseline at each horizon.

NEXT UP

Section 10: clustering and anomaly detection.

SECTION 10

10

Clustering and anomaly detection

Evaluating models when there is little or no ground truth to compare against.

IN THIS SECTION

- The challenge without labels
- Internal cluster measures
- External cluster measures
- Evaluating anomaly detection
- Human judgement

SECTION 10 · UNSUPERVISED

The challenge without labels



When there is no right answer to compare against, evaluation must rely on structure, stability and usefulness.

Structure

Are groups compact inside and well separated from each other?

Stability

Do similar clusters appear when the data or random seed changes?

Usefulness

Do the clusters or anomalies help people make better decisions?

SECTION 10 · UNSUPERVISED

Internal cluster measures

1 Silhouette score

How close each point is to its own cluster versus the nearest other cluster.

2 Davies Bouldin index

Average similarity between each cluster and its most similar neighbour. Lower is better.

3 Calinski Harabasz

Ratio of separation between clusters to spread within them. Higher is better.

4 Inertia

Total spread within clusters. Used in the elbow method for choosing cluster counts.

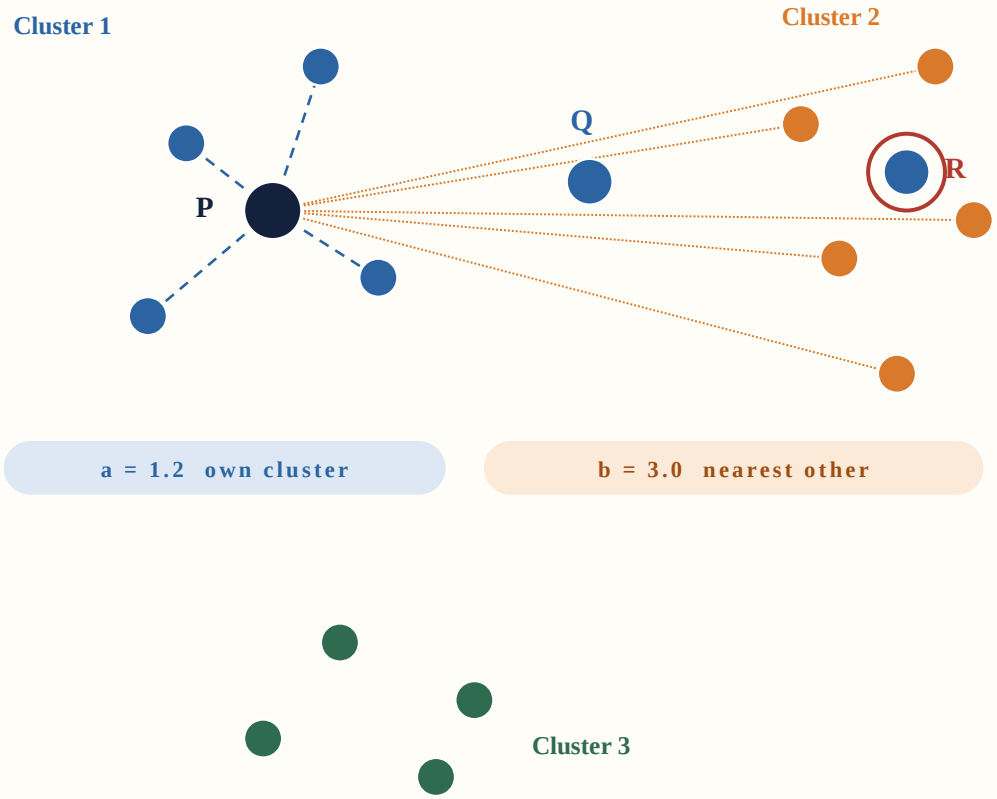
5 Stability

Agreement between clusterings on resampled data.

6 Size balance

Whether clusters are usefully sized rather than one giant group.

The silhouette score, computed for a single point

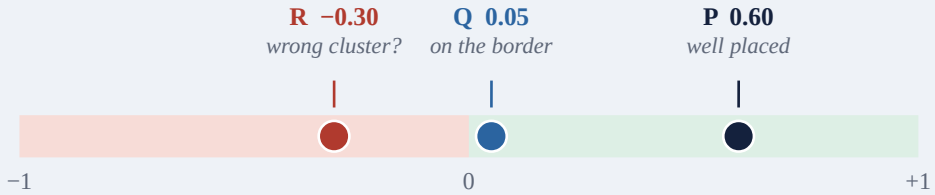


FOR POINT P

s = (b − a) ÷ max(a, b)

= (3.0 − 1.2) ÷ 3.0 = 0.60

WHERE EACH POINT LANDS



FROM POINTS TO A SCORE

The clustering's silhouette score is the average s over every point. Inspect the negative points: they are often mislabelled or sit between groups.

SECTION 10 · UNSUPERVISED

External cluster measures

When some reference labels exist, you can compare clusters against them, while remembering clusters may reveal structure the labels miss.

Adjusted Rand index

Agreement between clustering and labels, corrected for chance.

Mutual information

How much knowing the cluster tells you about the label.

Purity

Share of each cluster made up of its most common label.

SECTION 10 · UNSUPERVISED

Evaluating anomaly detection

Anomalies are rare and often unlabelled, so evaluation combines limited labels, injected examples and expert review.

Labelled incidents

Use known past incidents to measure precision and recall at the top alerts.

Synthetic anomalies

Inject realistic faults into clean data and check how many are caught.

Alert review

Have experts label a sample of alerts to estimate precision over time.

SECTION 10 · UNSUPERVISED

A practical review for unsupervised results

1. **Profile clusters.** Describe each cluster with summary statistics and examples.
2. **Name them.** Ask domain experts whether clusters have a meaningful interpretation.
3. **Test stability.** Rerun with different seeds and samples and compare.
4. **Check actionability.** Confirm each cluster suggests a different action.
5. **Track over time.** See whether clusters persist as new data arrives.

Unsupervised evaluation pitfalls

PITFALL	BETTER PRACTICE
⚠ Choosing cluster count by one internal score.	✓ Combine scores, stability and expert usefulness.
⚠ Treating every anomaly flag as a true problem.	✓ Measure alert precision with expert labels.
⚠ Ignoring feature scaling effects.	✓ Evaluate clusters under consistent preprocessing.
⚠ Clusters nobody can explain or use.	✓ Judge clusters by the decisions they enable.

SECTION 10 · UNSUPERVISED

Section recap

KEY TAKEAWAYS

- Unsupervised evaluation relies on structure, stability and usefulness
- Silhouette and related scores measure compactness and separation
- External measures compare with partial labels
- Anomaly detectors need labelled incidents and expert review
- Usefulness to people is the final test

TRY THIS

Cluster a dataset with three different cluster counts and compare silhouette scores, stability and how easily you can name each cluster.

NEXT UP

Section 11: language, vision and speech tasks.

SECTION 11

11

Language, vision and speech

Task specific metrics for unstructured data, and their important blind spots.

IN THIS SECTION

- Text classification and extraction
- Translation and summarisation
- Object detection
- Segmentation
- Speech recognition
- Human evaluation

Metrics by task

Task	Common metrics	Key caution
Text classification	Precision, recall, macro F1	Label noise in text data
Named entity recognition	Entity level precision and recall	Partial matches matter
Translation	BLEU, chrF, COMET	Overlap is not meaning
Summarisation	ROUGE, factual consistency	Fluent summaries can be false
Image classification	Top 1 and top 5 accuracy	Background shortcuts
Object detection	IoU, mAP	Thresholds change results
Speech recognition	Word error rate	Accents and noise vary

Overlap metrics and their limits

BLEU and ROUGE count words shared with reference texts. They are cheap and repeatable but can reward wrong answers that use the right words.

Strength

Fast, automatic and useful for tracking progress on one benchmark.

Weakness

Correct paraphrases score low and fluent mistakes can score high.

Better options

Learned metrics such as COMET or BERTScore, plus human review.

Intersection over union

DEFINITION

Intersection over union compares a predicted box or mask with the true one by dividing their overlapping area by the area they cover together.

EXAMPLE

A predicted box covering most of a car and a little road has high overlap with the true box and might score 0.8, counting as a correct detection at a 0.5 threshold.

REMEMBER

Detection results depend heavily on the overlap threshold chosen.

Intersection over union, drawn to scale

truth



IoU 0.76

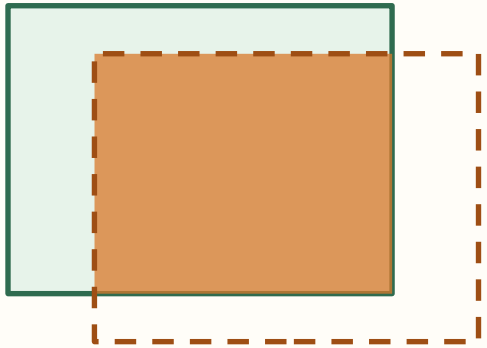
Good localisation

10.36 ÷ 13.64 units²

√ ≥ 0.50

√ ≥ 0.75

truth



IoU 0.48

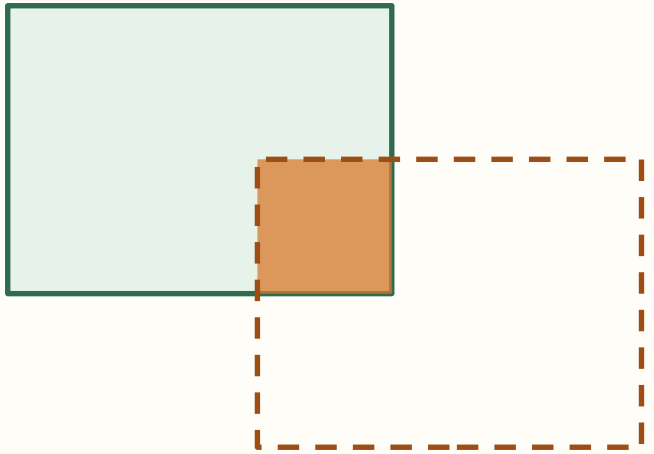
Just misses 0.5

7.75 ÷ 16.25 units²

✗ ≥ 0.50

✗ ≥ 0.75

truth



IoU 0.09

Clear miss

1.96 ÷ 22.04 units²

✗ ≥ 0.50

✗ ≥ 0.75

IoU =  overlap area ÷  area covered by both boxes together

! Averaging over thresholds 0.50 to 0.95 counts the first box at only 6 of 10 thresholds.

Mean average precision for detection

Mean average precision summarises detection quality across classes and confidence levels, and is often averaged over several overlap thresholds.

Per class

Average precision is computed for each object class from its precision recall curve.

Across thresholds

Some benchmarks average results over overlap thresholds from 0.5 to 0.95.

Also check

Small object performance and false detections per image.

Word error rate

DEFINITION

Word error rate counts the words a speech recogniser substituted, deleted or inserted, as a share of the words in the correct transcript.

EXAMPLE

A ten word sentence transcribed with one wrong word and one missing word has a word error rate of 20 percent.

REMEMBER

Report it separately for accents, noise levels and speaking styles.

Counting word errors by lining up the transcripts

Correct transcript	please	book	a	table	for	two	people	at	seven	tonight
Recogniser heard	please	look	a	table	for	two	*(missing)*	at	seven	tonight
Operation	✓	SUB	✓	✓	✓	✓	DEL	✓	✓	✓

S 1 substitution + D 1 deletion + I 0 insertions ÷ N 10 reference words = WER 20% word error rate

SURPRISE 1 · WER CAN EXCEED 100%

Truth “yes”, heard “yes yes okay”: 2 insertions ÷ 1 word = 200%.

SURPRISE 2 · ERRORS ARE NOT EQUAL

“Transfer fifteen pounds” heard as “transfer fifty pounds” is one error, 33% WER, and the wrong payment.

Running good human evaluation

1. **Write clear guidelines.** Define each rating with examples of good and bad outputs.
2. **Train raters.** Calibrate raters on practice items before real scoring.
3. **Blind the comparison.** Hide which system produced each output and randomise order.
4. **Use multiple raters.** Collect several ratings per item and measure agreement.
5. **Sample wisely.** Cover common cases and hard edge cases.
6. **Analyse disagreements.** Disagreements often reveal unclear guidelines or ambiguous tasks.

Pitfalls for unstructured data

PITFALL	BETTER PRACTICE
⚠ Trusting BLEU or ROUGE as quality.	→ ✓ Add learned metrics and human evaluation.
⚠ One overall word error rate.	→ ✓ Break results down by speaker group and condition.
⚠ Images with background shortcuts.	→ ✓ Test on images with changed backgrounds.
⚠ Benchmarks leaked into pretraining data.	→ ✓ Use fresh, private test sets.

Section recap

KEY TAKEAWAYS

- Each task has standard metrics and blind spots
- Overlap metrics measure words, not meaning
- IoU and mAP depend on thresholds
- Word error rate needs breakdowns by group
- Human evaluation needs guidelines, blinding and agreement checks

TRY THIS

Pick one text task and compare an overlap metric with your own human ratings on 30 examples. Where do they disagree?

NEXT UP

Section 12: evaluating large language models and generative systems.

SECTION 12

12

Evaluating LLMs and generative models

Measuring quality when there are many good answers and some dangerous ones.

IN THIS SECTION

- Why generation is hard to score
- Benchmarks and contamination
- Building golden sets
- LLM as a judge
- Evaluating RAG and agents
- Safety and red teaming

Why generative evaluation is hard



There is rarely one correct answer, and a fluent, confident response can still be completely wrong.

Open ended

Many different responses can be equally good.

Multi dimensional

Correctness, helpfulness, tone, safety and format all matter.

Moving target

Prompts, models and user needs change frequently.

Public benchmarks and their limits

Public benchmarks offer a shared yardstick for knowledge, reasoning and coding, but they rarely reflect your exact use case.

Useful for

Shortlisting models and tracking broad capability trends.

Contamination

Test questions may have appeared in training data, inflating scores.

Saturation

Top models cluster near the ceiling, so small differences mean little.

Building a golden evaluation set

1. **Collect real tasks.** Sample real user requests or realistic scenarios from your domain.
2. **Cover the range.** Include common, rare, ambiguous and adversarial cases.
3. **Define success.** Write reference answers or rubrics for each item.
4. **Review with experts.** Have domain experts check references and rubrics.
5. **Version it.** Store the set like code and record changes.
6. **Keep it private.** Protect it from leaking into training or prompt examples.

Ways to score generated output

1 Exact checks
Match against a known answer, number or label.

2 Programmatic tests
Run generated code against unit tests, report pass rates.

3 Format validation
Check JSON or schema compliance automatically.

4 Reference similarity
Semantic similarity to a reference answer.

5 LLM as a judge
A strong model scores outputs against a rubric.

6 Human rating
Experts score samples, the most trusted signal.

Using an LLM as a judge

A capable model can grade outputs at scale against a clear rubric, but its judgements must themselves be evaluated against human ratings.

Make it reliable

Give precise rubrics, examples and ask for reasoning before the score.

Known biases

Judges may favour longer answers, the first option or their own style.

Validate

Check agreement with expert ratings on a sample before trusting it.

Anatomy of a trustworthy LLM judge

```
judge_prompt.txt

ROLE   You grade customer-support answers.

RUBRIC 1. Facts correct ............ 0–2

       2. Answers the real question  0–2

       3. Concise, no padding ...... 0–1

FIRST  Explain your reasoning in 3 lines.

THEN   Output {"score": 0–5} and nothing else.
```

Precise criteria with point values

Reasoning before the score

Structured, parseable output

KNOWN BIASES AND THEIR FIXES

Position bias

Prefers whichever answer comes first.

Fix: Run both orders; count a win only if it survives the swap.

Verbosity bias

Rewards longer answers.

Fix: Credit concision in the rubric; compare at similar lengths.

Self-preference

Favours text in its own style.

Fix: Use a judge from another model family; hide the source.

VALIDATE THE JUDGE BEFORE YOU TRUST IT

1. Experts grade a sample of 200 outputs
2. The judge grades the same 200
3. Compare agreement, e.g. Cohen's kappa
4. Scale up only if it rivals expert-to-expert agreement

Evaluating a RAG system

Retrieval recall	Did the relevant passages appear in the retrieved set?
Retrieval precision	How much retrieved text was actually relevant?
Faithfulness	Is every claim in the answer supported by retrieved text?
Answer relevance	Does the answer address the question asked?
Citation quality	Do cited sources really support the statements?

Diagnose by stage

When a RAG answer is wrong, find out whether retrieval missed the evidence or generation misused it. The fixes are completely different.

Scoring one RAG answer at every stage of the pipeline



Retrieval recall	Retrieval precision	Faithfulness	Answer relevance	Citation quality
relevant passage found?	*how much is relevant?*	*every claim supported?*	*addresses the question?*	*sources back the claims?*

QUESTION **What is the refund window for online orders?**

P1	Refunds are accepted within 30 days of delivery.	*relevant*
P2	Our stores open 9am to 6pm, Monday to Saturday.	*noise*
P3	Gift cards cannot be exchanged for cash.	*noise*
P4	Standard delivery takes 3 to 5 working days.	*noise*
P5	The company was founded in 1998 in Leeds.	*noise*

GENERATED ANSWER

“You can return items within 30 days of delivery [P1], **and return shipping is always free** [P1].”

Claim 1 supported by P1

Claim 2 found in no passage

Diagnosis: search did its job. The generator invented a policy and cited P1 for it. Fix the prompt or the model, not the retriever.

100% Retrieval recall	**20%** Retrieval precision	**50%** Faithfulness	**Yes** Answer relevance	**50%** Citation quality
1 of 1	*1 of 5*	*1 of 2 claims*	*on topic*	*1 of 2 citations*

Evaluating agents

Agents plan and take actions with tools across many steps, so evaluation must look at outcomes, paths, cost and safety together.

Task success

Did the agent achieve the goal in a sandboxed, realistic environment?

Trajectory

Were the steps sensible, efficient and free of harmful actions?

Cost and time

How many steps, tokens and seconds did success require?

Safety evaluation and red teaming

1 Harmful content
Test refusal of dangerous requests and absence of harmful output.

2 Prompt injection
Test whether hidden instructions in documents hijack behaviour.

3 Privacy
Check that personal or confidential data is not leaked.

4 Hallucination
Measure how often unsupported claims appear.

5 Bias
Compare responses across names, groups and dialects.

6 Over refusal
Make sure safe requests are not refused unnecessarily.

Practical LLM quality signals

Signal	What it tells you
Pass rate on golden set	Core task quality against agreed answers
Judge score by rubric	Scalable quality across several dimensions
Faithfulness rate	How often answers stay grounded in sources
Refusal rates	Balance between safety and helpfulness
User feedback	Thumbs, edits and follow up questions in production
Cost and latency	Whether quality is affordable at scale

Variance in generative evaluation

The same prompt can produce different outputs each run, so a single run per item gives noisy, unreliable comparisons.

Repeat runs

Run each item several times and report the average and spread.

Fix settings

Record temperature, model version and prompt version for every run.

Paired comparisons

Compare systems on the same items to reduce noise.

LLM evaluation pitfalls

PITFALL	BETTER PRACTICE
⚠ Deciding by vibes from a few demos.	✓ Use a versioned golden set and rubrics.
⚠ Trusting an unvalidated LLM judge.	✓ Measure judge agreement with experts first.
⚠ Leaderboard scores as proof of fitness.	✓ Test on your own tasks and data.
⚠ Ignoring retrieval when RAG answers fail.	✓ Score retrieval and generation separately.

Section recap

KEY TAKEAWAYS

- Generative outputs need multi dimensional evaluation
- Benchmarks shortlist, golden sets decide
- LLM judges scale but need human validation
- Evaluate RAG retrieval and faithfulness separately
- Agents need outcome, trajectory and safety checks
- Repeat runs to handle output variance

TRY THIS

Create a 25 item golden set with rubrics for an LLM task, score two prompts, and validate an LLM judge against your own ratings.

NEXT UP

Section 13: statistical rigour in evaluation.

SECTION 13

13

Statistical rigour

Knowing whether a difference between models is real or just luck.

IN THIS SECTION

- Every metric is an estimate
- Confidence intervals
- The bootstrap
- Comparing two models
- Multiple comparisons
- Seeds and variance

A metric without an uncertainty estimate is half a result.

☆ Test sets are samples. A different sample would give a slightly different number, and decisions must account for that.

Confidence intervals

DEFINITION

A confidence interval gives a range of plausible values for a metric, reflecting how much it could vary if you evaluated on another sample of the same size.

EXAMPLE

A model’s recall is 82 percent with a 95 percent interval from 78 to 86 percent. A rival at 84 percent is not clearly better.

REMEMBER

Overlapping intervals are a warning that differences may be noise.

SECTION 13 · STATISTICAL RIGOUR

The bootstrap, step by step

01

Resample

Draw a sample of test rows with replacement, the same size as the test set.

02

Score

Compute the metric on that resampled set.

03

Repeat

Do this a thousand times or more.

04

Sort

Line up all the resulting metric values.

05

Read off

The middle 95 percent forms the confidence interval.

The bootstrap works for almost any metric, including F1, AUC and NDCG, without needing special formulas.

The bootstrap, watched resample by resample

ORIGINAL TEST SET: 10 ROWS

	1	2	3	4	5	6	7	8	9	10	
Original	✓	✓	✗	✓	✓	✓	✗	✓	✗	✓	70%

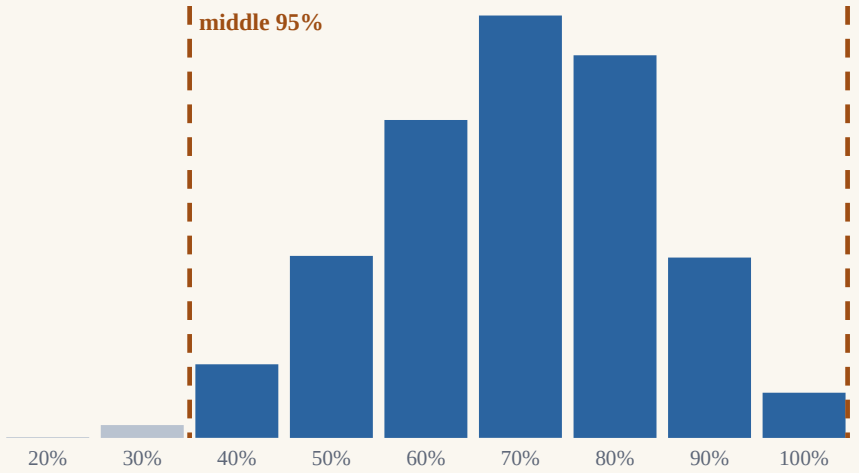
DRAW 10 ROWS WITH REPLACEMENT, AGAIN AND AGAIN

Resample 1	9 ✗	1 ✓	2 ✓	3 ✗	2 ✓	9 ✗	9 ✗	6 ✓	1 ✓	1 ✓	60%
Resample 2	4 ✓	5 ✓	7 ✗	5 ✓	3 ✗	2 ✓	7 ✗	8 ✓	1 ✓	2 ✓	70%
Resample 3	5 ✓	4 ✓	9 ✗	6 ✓	5 ✓	5 ✓	7 ✗	6 ✓	2 ✓	8 ✓	80%

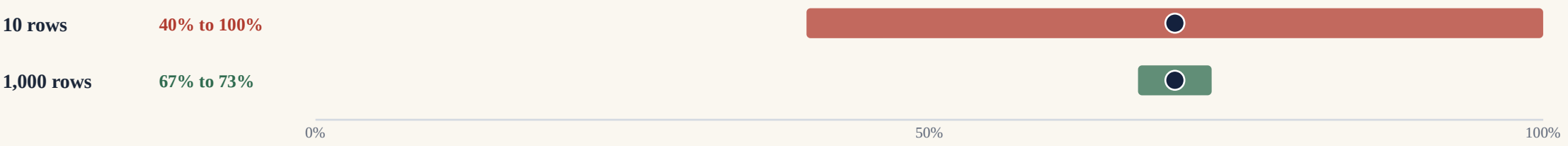
Dark outline = a row drawn more than once.

⋮ *… 2,000 resamples in total*

2,000 RESAMPLED ACCURACIES



SAME 70% ACCURACY, DIFFERENT CERTAINTY



Bootstrap interval for any metric

bootstrap_ci.py

```
import numpy as np
from sklearn.metrics import recall_score

rng = np.random.default_rng(7)
n = len(y_true)
stats = []
for _ in range(2000):
    idx = rng.integers(0, n, n)
    stats.append(recall_score(y_true[idx], y_pred[idx]))
low, high = np.percentile(stats, [2.5, 97.5])
print(f"recall interval: {low:.3f} to {high:.3f}")
```

Any metric

Swap recall for F1, AUC or a custom business metric.

Keep pairs

Resample predictions and labels together, row by row.

Groups

If rows cluster by user, resample whole users instead.

Tests for comparing two models

Test	Compares	Typical use
McNemar’s test	Disagreements on the same items	Two classifiers, one test set
Paired bootstrap	Metric difference on resamples	Almost any metric
Paired t test	Average difference across folds or items	Roughly normal differences
Wilcoxon signed rank	Ranks of paired differences	No normality assumption
DeLong test	Two ROC AUC values	Comparing AUCs on same data

Always compare models on the same test items. Paired comparisons remove much of the noise from item difficulty.

Statistical versus practical significance

STATISTICAL

Is the difference real?

- Asks whether chance could explain it
- Huge test sets make tiny gains significant
- Reported with p values or intervals
- Necessary but not sufficient

PRACTICAL

Does the difference matter?

- Asks whether it changes outcomes
- Judged against costs and effort
- Expressed as an effect size in business terms
- Decides whether to ship

VERDICT Ship only when a difference is both statistically credible and large enough to justify its cost and risk.

SECTION 13 · STATISTICAL RIGOUR

The multiple comparisons trap

Test enough models, slices or metrics and some will look better purely by chance. The more you look, the more false discoveries you make.

Example

Comparing 20 model variants at a 5 percent threshold will often produce one false winner.

Corrections

Bonferroni and Holm methods tighten thresholds when many tests are run.

Better habit

Pre register the main comparison and confirm surprises on fresh data.

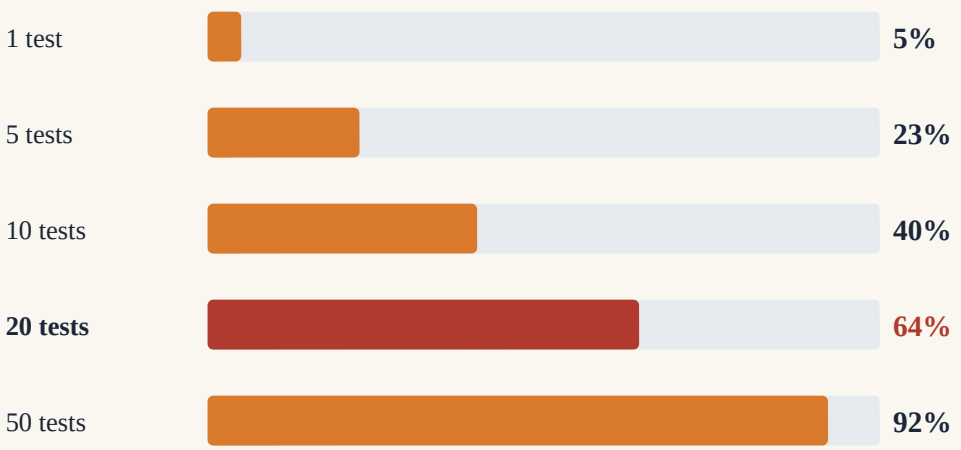
Twenty variants, no real winner, one false discovery

20 VARIANTS VS BASELINE · NONE TRULY BETTER

Variant 1 **p = 0.41**	Variant 2 **p = 0.77**	Variant 3 **p = 0.18**	Variant 4 **p = 0.62**	Variant 5 **p = 0.09**
Variant 6 **p = 0.55**	Variant 7 **p = 0.93**	Variant 8 **p = 0.27**	Variant 9 **p = 0.70**	Variant 10 **p = 0.36**
Variant 11 **p = 0.12**	Variant 12 **p = 0.84**	Variant 13 **p = 0.48**	**Variant 14 p = 0.03**	Variant 15 **p = 0.66**
Variant 16 **p = 0.22**	Variant 17 **p = 0.59**	Variant 18 **p = 0.31**	Variant 19 **p = 0.88**	Variant 20 **p = 0.15**

Variant 14 looks "significant". *It is pure luck: on noise alone, about 1 in 20 tests passes.*

CHANCE OF AT LEAST ONE FALSE WIN AT P < 0.05



$1 - 0.95^n$, assuming independent tests

BONFERRONI CORRECTION

Threshold 0.05 ÷ 20 = 0.0025. Variant 14 (p = 0.03) **no longer passes.**

HABIT Name the one comparison that matters before you look, then confirm any surprise on fresh data.

SECTION 13 · STATISTICAL RIGOUR

Variance from seeds and training

Random initialisation, data order and sampling make results vary between training runs, sometimes as much as the improvement being claimed.

Run several seeds

Train with at least three to five seeds and report the spread.

Compare distributions

Judge whether one method’s range sits clearly above the other’s.

Report honestly

State how many runs were done and how many were discarded.

SECTION 13 · STATISTICAL RIGOUR

How big should a test set be?

1 Rare events dominate

Precision on rare classes needs enough positives, not just rows.

2 Smaller effects need more data

Detecting a one point gain needs far more items than ten points.

3 Use power analysis

Estimate sample size before collecting labels.

4 Check interval width

If intervals are wider than the gain you care about, get more data.

5 Stratify sampling

Oversample important slices for labelling, then reweight.

6 Reuse carefully

Repeated use of one test set slowly overfits it.

Statistical pitfalls

PITFALL	BETTER PRACTICE
⚠ Declaring a winner from a 0.3 point gain.	✓ Report intervals and paired tests.
⚠ Testing many variants, reporting the best.	✓ Correct for multiple comparisons and confirm on fresh data.
⚠ One training run per method.	✓ Run several seeds and show the spread.
⚠ Bootstrapping rows when users repeat.	✓ Resample at the level of independent units.

Section recap

KEY TAKEAWAYS

- Every metric is an estimate with uncertainty
- Confidence intervals show the plausible range
- The bootstrap works for almost any metric
- Use paired tests on the same items
- Separate statistical from practical significance
- Correct for many comparisons and run several seeds

TRY THIS

Compute bootstrap intervals for two models' main metric and a paired bootstrap of their difference. Is the winner really ahead?

NEXT UP

Section 14: error analysis and slicing.

SECTION 14

14

Error analysis and slicing

Moving from how good is the model to where, why and for whom it fails.

IN THIS SECTION

- Why aggregates hide failures
- Slice based evaluation
- Reading the errors
- Error taxonomies
- Behavioural testing
- Prioritising fixes

Averages hide failures



A model can improve on average while getting worse for an important group. Slicing makes those hidden failures visible.

Hidden stratification

Rare but important subgroups vanish in overall metrics.

Simpson’s paradox

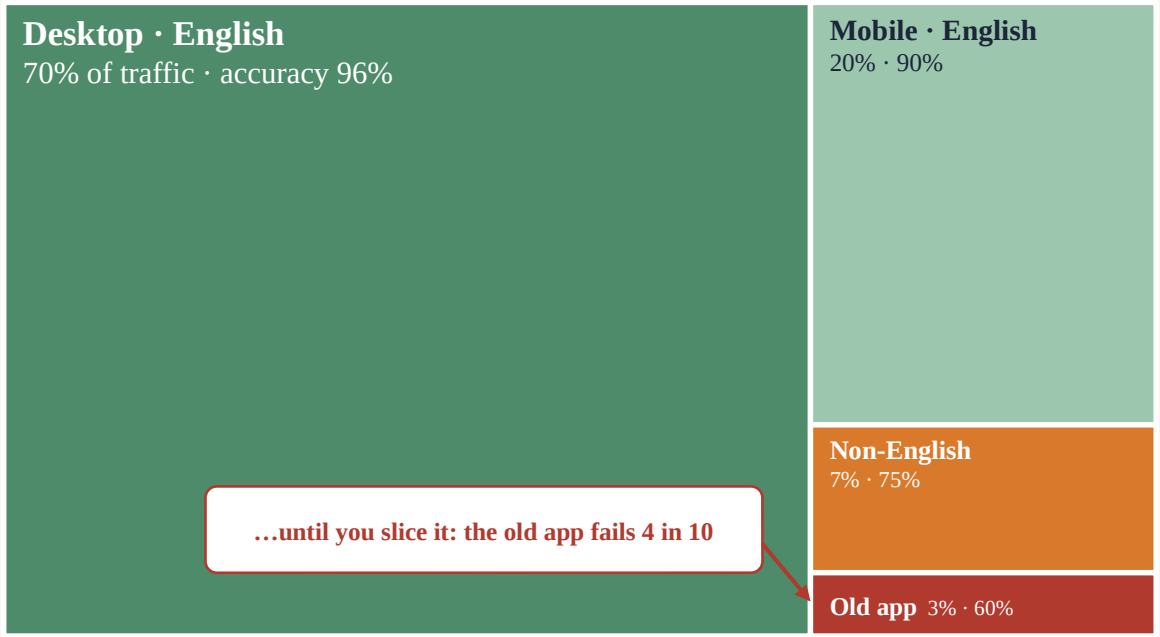
A model can win on every slice yet lose overall, or the reverse.

Business impact

Key customers, products or regions deserve their own metrics.

A 92% average hiding a 60% slice, and Simpson's paradox

Overall accuracy **92%** *looks great…*



Area = share of traffic · colour = accuracy

SIMPSON'S PARADOX IN A MODEL COMPARISON

Slice	Model A	Model B
Easy cases	270 / 300 = 90%	**92 / 100 = 92%** ✓
Hard cases	35 / 100 = 35%	**120 / 300 = 40%** ✓
Overall	**305 / 400 = 76%** ✓	212 / 400 = 53%

WHY IT FLIPS

B wins on both slices yet loses overall, because B was scored on 300 hard cases and A on only 100. The overall figure measures the case mix, not the model.

! Score both models on the same test set, and report key slices beside the average.

Useful ways to slice

1 Demographics

Age bands, regions, languages, where lawful and appropriate.

2 Data source

Channel, device, sensor, partner or upstream system.

3 Time

Hour, weekday, season and recent versus older data.

4 Input traits

Length, image quality, missing fields, rare categories.

5 Confidence

High, medium and low model confidence bands.

6 Business value

High value customers, critical products, priority cases.

SECTION 14 · ERROR ANALYSIS

A systematic error analysis

1. **Collect errors.** Gather a sample of false positives and false negatives.
2. **Read them.** Look at the raw records, text or images, not just numbers.
3. **Tag causes.** Label each error with one or more likely causes.
4. **Count categories.** Find which causes account for most errors.
5. **Estimate gains.** Estimate how much fixing each category would help.
6. **Fix and recheck.** Address the biggest category and evaluate again.

An example error taxonomy

Error category	Example	Typical fix
Label noise	Training label was simply wrong	Relabel and audit labels
Missing feature	Key information not in inputs	Add or engineer features
Rare pattern	Too few examples of this case	Collect or augment data
Ambiguous case	Even experts disagree	Clarify guidelines or abstain
Distribution shift	Inputs differ from training	Retrain on recent data
Model limitation	Pattern too complex for the model	Change architecture

Behavioural testing

Behavioural tests check capabilities directly with targeted examples, like unit tests for model behaviour.

Invariance

Changing a name or irrelevant word should not change the prediction.

Directional

Adding a clearly negative phrase should lower a sentiment score.

Minimum functionality

Simple, obvious cases must always be correct.

SECTION 14 · ERROR ANALYSIS

Prioritising what to fix



Quick wins

Many errors, little effort. Fix these first.

Major projects

Many errors, large effort. Plan and resource them.

Small polish

Few errors, little effort. Batch them together.

Defer

Few errors, large effort. Revisit later if impact grows.

Vertical axis: Share of errors (high at top)

Horizontal axis: Effort to fix (low at left)

SECTION 14 · ERROR ANALYSIS

Error analysis pitfalls

PITFALL	BETTER PRACTICE
⚠ Only looking at aggregate metrics.	✓ Read real errors and slice results.
⚠ Slices with tiny sample sizes.	✓ Show counts and intervals beside slice metrics.
⚠ Fixing the most interesting errors first.	✓ Prioritise by frequency, impact and effort.
⚠ One off analysis that is never repeated.	✓ Automate slice reports for every candidate model.

Section recap

KEY TAKEAWAYS

- Aggregates hide important failures
- Slice by group, source, time, input traits and value
- Read errors and tag their causes
- Use a taxonomy to count and prioritise
- Behavioural tests check specific capabilities
- Automate slice reports in your pipeline

TRY THIS

Read 50 errors from your model, tag their causes and present the top three categories with estimated gains.

NEXT UP

Section 15: fairness evaluation.

SECTION 15

15

Fairness evaluation

Checking that a model's benefits and harms are not unfairly concentrated on some people.

IN THIS SECTION

- Sources of unfairness
- Group fairness definitions
- Measuring disparities
- Impossibility results
- Individual fairness
- Documenting findings

SECTION 15 · FAIRNESS

Where unfairness comes from

1 Historical bias
Past decisions in the data reflect past discrimination.

2 Representation
Some groups have far fewer training examples.

3 Measurement
Labels or features are less accurate for some groups.

4 Proxies
Postcodes or names stand in for protected characteristics.

5 Aggregation
One model fits the majority and fails smaller groups.

6 Deployment
The model is used in ways or places it was not built for.

Common group fairness definitions

Definition	Idea in plain words	Concern it addresses
Demographic parity	Similar positive rates across groups	Unequal access to outcomes
Equal opportunity	Similar recall across groups	Qualified people missed
Equalised odds	Similar recall and false alarm rates	Unequal error burdens
Predictive parity	Similar precision across groups	Unequal meaning of a positive
Calibration by group	Scores mean the same in every group	Misleading risk scores

Which definition fits depends on context, law and stakeholder values. Choose deliberately and document why.

SECTION 15 · FAIRNESS

You cannot satisfy every definition at once

⚠ Research has shown that when base rates differ between groups, several fairness definitions are mathematically incompatible except in special cases.

Implication

Some trade off is unavoidable, so fairness is a decision, not just a calculation.

Involve people

Affected communities, legal and domain experts should help choose priorities.

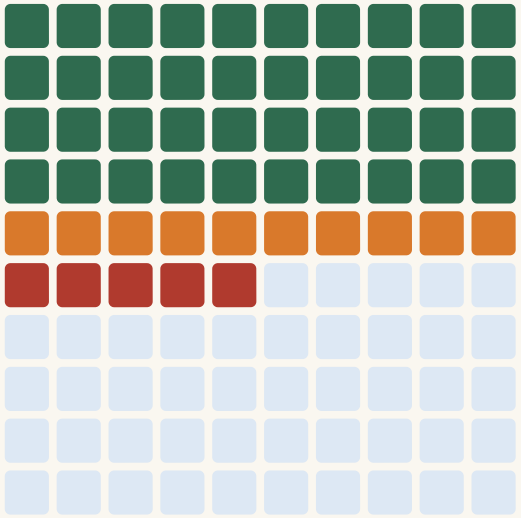
Be transparent

Report which definitions you met and which you did not.

Why fairness definitions collide: one model, two groups

Group A

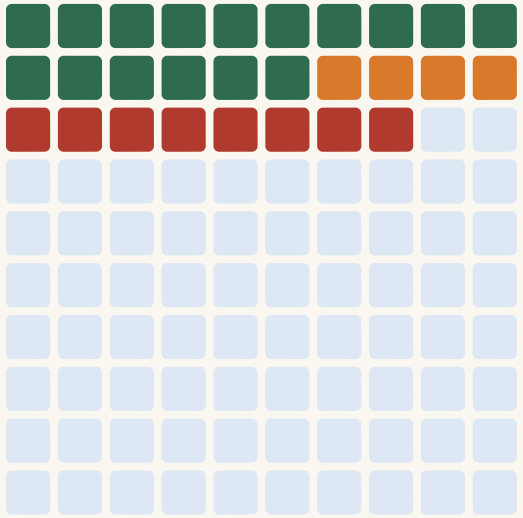
1,000 people · base rate 50%



TP 400 · FN 100 · FP 50 · TN 450

Group B

1,000 people · base rate 20%



TP 160 · FN 40 · FP 80 · TN 720

Caught (TP)

False alarm (FP)

Missed (FN)

Correctly cleared (TN)

Each square = 10 people

SCORECARD

Measure	A	B	Definition
True positive rate	80%	80%	✓ Equal opportunity
False positive rate	10%	10%	✓ Equalised odds
Precision	89%	67%	✗ Predictive parity
Selection rate	45%	24%	✗ Demographic parity

THE COLLISION

Raise B's precision to 89% and you must change its threshold, which moves its error rates away from A's. With different base rates, something has to give.

Choosing which one gives is a decision for people, not a calculation.

SECTION 15 · FAIRNESS

Measuring disparities in practice

1. **Define groups.** Identify relevant groups lawfully and with appropriate safeguards.
2. **Pick metrics.** Choose fairness metrics linked to the harms that matter.
3. **Compute per group.** Calculate metrics for each group with confidence intervals.
4. **Compare gaps.** Report differences or ratios between groups.
5. **Check intersections.** Look at combinations, such as age within gender.
6. **Set tolerances.** Agree acceptable gaps before deciding to ship.

SECTION 15 · FAIRNESS

Individual and counterfactual fairness

Group metrics can miss unfair treatment of individuals. Individual approaches ask whether similar people are treated similarly.

Similar treatment

Two applicants alike in relevant ways should receive similar scores.

Counterfactual tests

Change only a sensitive attribute or its proxy and check whether the decision flips.

Name swap tests

For text models, swap names associated with different groups and compare outputs.

Tools that support fairness evaluation

Tool	What it offers
Fairlearn	Group metrics, disparity dashboards and mitigation methods
AIF360	Large collection of fairness metrics and algorithms
What If Tool	Interactive exploration of model behaviour by slice
Cloud tooling	Bias reports in SageMaker Clarify, Vertex AI and Azure ML

SECTION 15 · FAIRNESS

Fairness pitfalls

PITFALL	BETTER PRACTICE
⚠ Dropping protected attributes and calling it fair.	→ ✓ Test for proxies and measure outcomes by group.
⚠ Reporting one fairness metric without context.	→ ✓ Explain the choice and show several metrics.
⚠ Ignoring small or intersectional groups.	→ ✓ Report counts and intervals for every group.
⚠ Fairness checked once at launch.	→ ✓ Monitor group metrics continuously in production.

SECTION 15 · FAIRNESS

Knowledge check

? A hiring screen has equal precision across two groups but much lower recall for one group. Is it fair?

MODEL ANSWER

It satisfies predictive parity but fails equal opportunity: qualified candidates from one group are missed far more often. Given the harm of being wrongly screened out, many would judge this unacceptable. The team should investigate causes and agree, with legal and stakeholder input, which criterion must hold.

WHY IT MATTERS

Different definitions can tell opposite stories about the same model.

SECTION 15 · FAIRNESS

Section recap

KEY TAKEAWAYS

- Unfairness enters through data, labels, proxies and deployment
- Group definitions measure different kinds of harm
- Several definitions cannot all hold at once
- Measure per group with intervals and intersections
- Test individuals with counterfactual and name swap checks
- Monitor fairness after launch, not only before

TRY THIS

Compute recall, false positive rate and precision per group for one model and write a short paragraph on which gaps matter and why.

NEXT UP

Section 16: robustness and uncertainty.

SECTION 16

16

Robustness and uncertainty

Evaluating how models behave when the world does not look like the test set.

IN THIS SECTION

- Distribution shift
- Perturbation tests
- Adversarial robustness
- Out of distribution detection
- Uncertainty estimation
- Selective prediction

SECTION 16 · ROBUSTNESS

Testing under distribution shift

Real data drifts away from training data. Robustness evaluation deliberately tests on data that differs in realistic ways.

Temporal shift

Evaluate on later periods than training.

Domain shift

Evaluate on new regions, hospitals, devices or customer types.

Report the drop

Show how much performance falls compared with the in distribution test.

Perturbation tests

1 **Noise**
Add realistic sensor, image or audio noise.

2 **Typos and casing**
Introduce spelling errors and capitalisation changes in text.

3 **Missing values**
Remove fields that may be missing in production.

4 **Unit and format changes**
Feed dates, currencies or units in new formats.

5 **Image changes**
Blur, rotate, crop and change lighting.

6 **Paraphrases**
Rephrase prompts or questions while keeping their meaning.

SECTION 16 · ROBUSTNESS

Adversarial robustness

Adversarial evaluation searches for small, deliberate input changes that flip predictions, revealing how easily an attacker could fool the model.

Why it matters

Fraud, spam, security and content moderation face active adversaries.

How to test

Use attack libraries or human red teams to find failing inputs.

Report

Accuracy under attack at a stated attack strength, not only clean accuracy.

Out of distribution detection

DEFINITION

Out of distribution detection evaluates whether a model can recognise inputs unlike anything it was trained on, so it can flag them instead of guessing confidently.

EXAMPLE

A skin lesion classifier receives a photo of a pet. A good system flags it as unfamiliar rather than confidently predicting a diagnosis.

REMEMBER

Evaluate how well the system separates familiar from unfamiliar inputs.

SECTION 16 · ROBUSTNESS

Evaluating uncertainty estimates

Uncertainty is useful only if it is meaningful: high uncertainty should go with more errors, and stated confidence should match reality.

Calibration

Confidence levels should match observed accuracy.

Error correlation

Uncertain predictions should be wrong more often than confident ones.

Methods

Ensembles, Monte Carlo dropout and conformal prediction produce uncertainty.

SECTION 16 · ROBUSTNESS

Conformal prediction

DEFINITION

Conformal prediction wraps any model to output a set or range of answers with a stated coverage guarantee, under the assumption that new data resembles the calibration data.

EXAMPLE

Instead of one diagnosis, a model returns the set pneumonia or bronchitis, built so that about 90 percent of such sets contain the true answer.

REMEMBER

Evaluate both coverage and the average size of the sets.

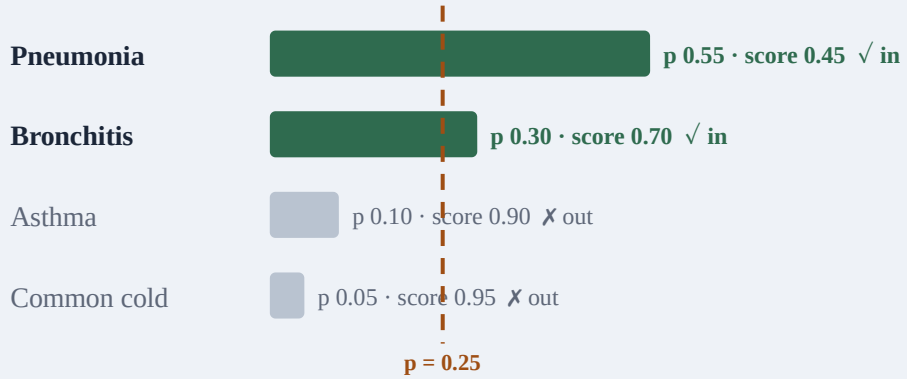
Conformal prediction: from calibration scores to a prediction set

1 **Calibration patients:** score = 1 − probability the model gave the true diagnosis, sorted

0.02	0.04	0.06	0.09	0.11	0.14	0.17	0.20	0.24	0.28	0.31	0.36	0.41	0.47	0.53	0.60	0.68	0.75	0.88
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	**18**	19

2 **Pick the cut-off:** for 90% coverage take rank $\lceil (19 + 1) \times 0.9 \rceil = 18$, so **q = 0.75**

3 **New patient:** keep every diagnosis with score ≤ 0.75, i.e. probability ≥ 0.25



PREDICTION SET

{ pneumonia, bronchitis }

SET SIZE SIGNALS UNCERTAINTY

Clear case → 1 label
Flu 0.93, others under 0.05

Murky case → 3 labels
0.30, 0.28, 0.26 all clear 0.25

! Guarantee: about 90% of sets contain the true diagnosis, provided new patients resemble the calibration patients.

SECTION 16 · ROBUSTNESS

Selective prediction and abstention

Letting a model abstain on uncertain cases and pass them to humans can make the automated portion far more reliable.

Coverage

Share of cases the model decides automatically.

Selective accuracy

Accuracy on the cases it chooses to decide.

Trade off curve

Plot accuracy against coverage to choose an operating point.

Robustness pitfalls

PITFALL	BETTER PRACTICE
⚠ Testing only on clean, curated data.	✓ Add realistic noise, shift and missing data tests.
⚠ Trusting raw confidence scores.	✓ Evaluate calibration and uncertainty quality.
⚠ Ignoring active adversaries.	✓ Red team models used in adversarial settings.
⚠ No path for unfamiliar inputs.	✓ Add detection and human escalation.

Section recap

KEY TAKEAWAYS

- Test on realistic shifted data and report the drop
- Perturbation tests reveal brittleness
- Adversarial tests matter when attackers exist
- Detect unfamiliar inputs rather than guessing
- Uncertainty must be calibrated and correlated with errors
- Abstention trades coverage for reliability

TRY THIS

Apply three realistic perturbations to your test set and report how much each lowers your main metric.

NEXT UP

Section 17: interpretability as part of evaluation.

SECTION 17

17

Interpretability checks

Using explanations to confirm the model is right for the right reasons.

IN THIS SECTION

- Why explanations aid evaluation
- Global importance
- Local explanations
- Catching shortcuts
- Evaluating explanations

Right for the right reasons

Accuracy shows a model is right. Explanations help show whether it is right for reasons that will still hold tomorrow.

Find shortcuts

Reveal reliance on watermarks, IDs or leaked fields.

Build trust

Domain experts can sanity check what drives predictions.

Support decisions

Explain individual outcomes to users and reviewers.

SECTION 17 · INTERPRETABILITY

Global explanation methods

1 Permutation importance

How much performance drops when a feature is shuffled.

2 Mean SHAP values

Average contribution of each feature across many predictions.

3 Partial dependence

How predictions change as one feature varies.

4 Accumulated effects

Similar to partial dependence but safer with correlated features.

5 Surrogate models

A simple model that imitates the complex one for inspection.

6 Coefficients

Direct weights in linear and logistic models.

SECTION 17 · INTERPRETABILITY

Local explanations

Local methods explain one prediction at a time, showing which inputs pushed this specific result up or down.

SHAP

Shares out a prediction among features using ideas from cooperative game theory.

LIME

Fits a simple model around one example to approximate local behaviour.

Counterfactuals

Shows the smallest change that would flip the decision.

SECTION 17 · INTERPRETABILITY

Shortcut learning

DEFINITION

Shortcut learning happens when a model relies on an easy but irrelevant signal that happens to correlate with the label in training data.

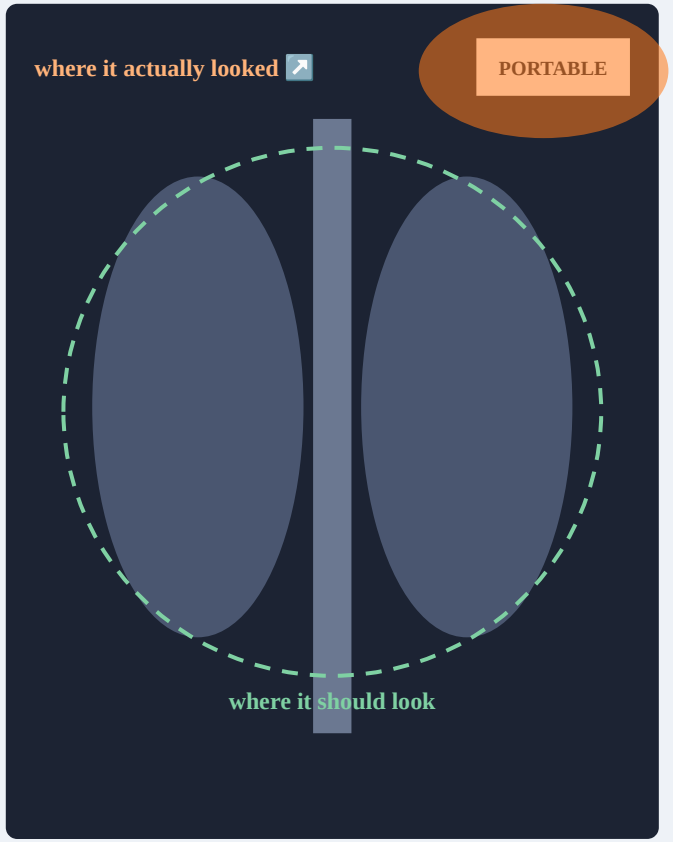
EXAMPLE

A chest scan model learned to detect a hospital's scanner marker, which was more common in sicker patients, instead of signs of disease.

REMEMBER

If explanations point at something an expert finds absurd, investigate before shipping.

Catching a shortcut: the scan model that read the label



Stylised scan with an explanation heat map

HOW THE TEAM PROVED IT

1 Explanation
Heat maps lit up the corner marker on most positive cases.

2 Mask the marker
Cover the corner and re-score the same test set.
AUC 0.93 → 0.78

3 New hospital
Score scans from a site with different scanners.
AUC 0.71

AUC values illustrative

SHORTCUTS SEEN ELSEWHERE

Wolf or husky?
Classifier keyed on snow in the background, not the animal.

Skin lesions
Rulers placed beside worrying lesions became a cue for malignancy.

Text tasks
Single words such as “not” predicted the label regardless of meaning.

TEST FOR IT
Mask, swap or shuffle the suspect signal. If scores collapse, you found a shortcut.

Interpretability pitfalls

PITFALL	BETTER PRACTICE
⚠ Treating explanations as ground truth.	✓ Check explanation stability and faithfulness.
⚠ Misreading importance as causation.	✓ Remember importance shows association, not cause.
⚠ Ignoring correlated features.	✓ Group correlated features or use robust methods.
⚠ Explaining only the averages.	✓ Inspect local explanations for errors and slices.

Section recap

KEY TAKEAWAYS

- Explanations help confirm the model uses sensible signals
- Global methods show overall drivers
- Local methods explain single predictions
- Shortcut learning is caught by expert review of explanations
- Explanations show association, not causation

TRY THIS

Generate permutation importance and five local explanations for your model. Ask a domain expert if anything looks wrong.

NEXT UP

Section 18: online evaluation and A/B testing.

SECTION 18

18

Online evaluation and A/B tests

Proving that a model creates real value for real users.

IN THIS SECTION

- Why online evaluation
- Designing an experiment
- Choosing metrics
- Sample size and duration
- Threats to validity
- Reading results

SECTION 18 · ONLINE EVALUATION

Why offline wins do not always transfer



Offline metrics measure predictions on historical data. Online tests measure what happens when predictions change behaviour.

Feedback effects

Users react to the model, creating data no offline test contained.

Proxy gaps

Better clicks offline may not mean more revenue or satisfaction.

System effects

Latency, interface and integration all shape real outcomes.

Designing an A/B test

1. **Hypothesis.** State what change you expect and why, in plain words.
2. **Unit.** Choose what is randomised, usually users rather than requests.
3. **Metrics.** Fix one primary metric, secondary metrics and guardrails.
4. **Size and duration.** Compute sample size and run full weekly cycles.
5. **Analysis plan.** Decide the statistical method and stopping rules upfront.
6. **Launch criteria.** Agree what result leads to ship, iterate or stop.

A hierarchy of experiment metrics

Primary metric	The single outcome the test is designed to move, such as conversion.
Secondary metrics	Supporting signals that explain how the change worked.
Guardrail metrics	Things that must not get worse, such as latency or complaints.
Model metrics	Online precision, coverage and prediction distributions.
Diagnostic metrics	Sample ratio, exposure counts and data quality checks.

Keep it focused

One primary metric decides the outcome. Guardrails can veto a launch even when the primary metric improves.

SECTION 18 · ONLINE EVALUATION

Sample size and duration

1 **Minimum detectable effect**

The smallest improvement worth detecting drives sample size.

2 **Baseline rate**

Rare outcomes need much larger samples.

3 **Variance**

Noisy metrics like revenue need more users than click rates.

4 **Full cycles**

Run at least one or two full weeks to cover weekday patterns.

5 **Variance reduction**

Methods such as CUPED use pre experiment data to reduce noise.

6 **Power calculators**

Use standard calculators before launch, not after.

Threats to validity

1 Peeking
Stopping as soon as results look good inflates false wins.

2 Sample ratio mismatch
Unequal group sizes signal a broken assignment.

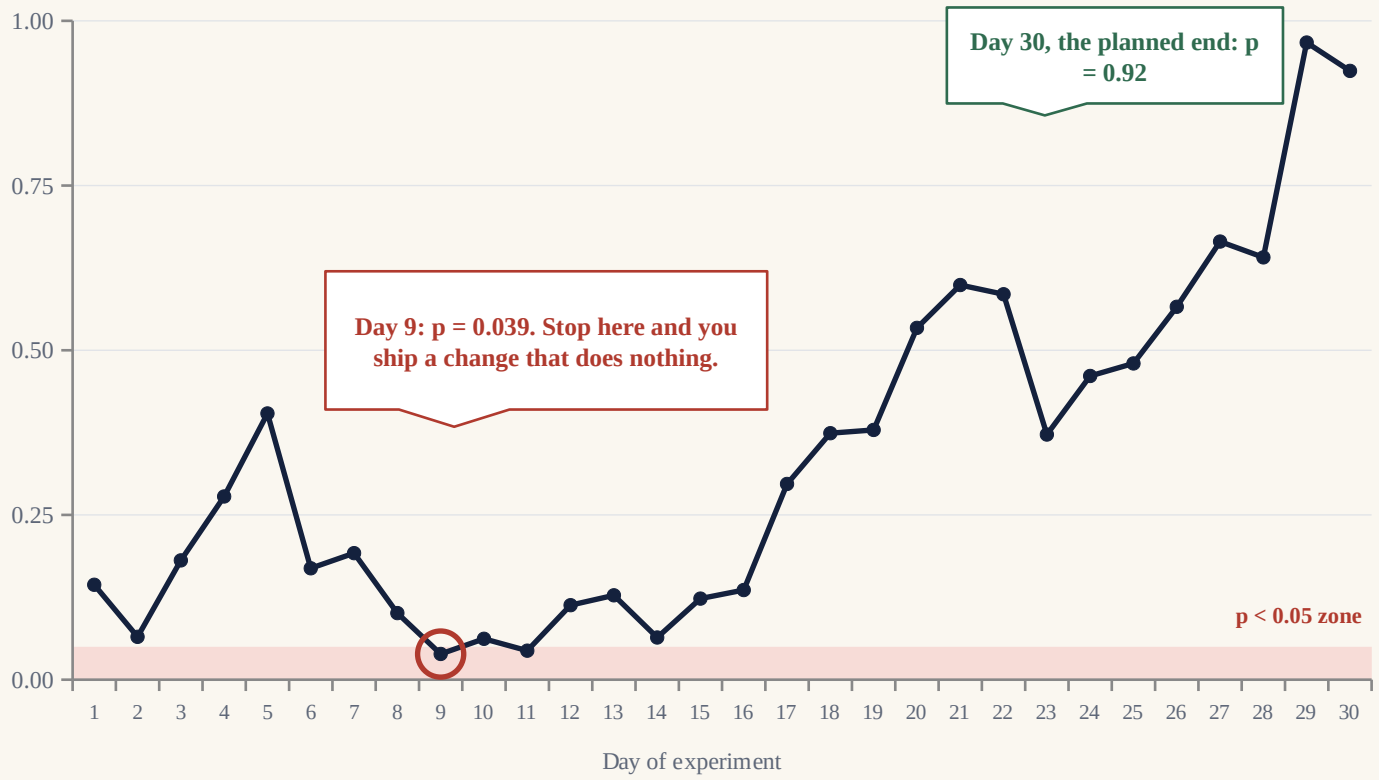
3 Novelty effect
Users react to change itself, then settle down.

4 Interference
Users in different groups affect each other, such as in marketplaces.

5 Multiple metrics
Checking many metrics produces chance wins.

6 Seasonality
Holidays or events during the test distort results.

Peeking: the p-value that wanders below 0.05 and back



Simulated A/A test: identical experience in both groups.

FALSE WIN RATE WITH DAILY PEEKING

≈ 27%

not 5%. From 4,000 simulated A/A tests checked daily for 30 days.

WAYS TO LOOK SAFELY

Fix it in advance

Set sample size and duration before launch; read once.

Use a sequential method

Tests designed for repeated looks keep the error rate honest.

Watch guardrails, not p

Monitor for harm daily; judge the primary metric at the end.

Reading A/B results responsibly

A good readout reports the effect size with its confidence interval, the guardrails and any validity checks, not just a p value.

Effect and interval

Conversion rose 1.2 percent, plausibly between 0.4 and 2.0 percent.

Guardrails

Latency, complaints and refunds stayed within limits.

Segments

Pre planned segment results, labelled exploratory if not planned.

SECTION 18 · ONLINE EVALUATION

A/B tests versus bandits for models

A/B TEST

Fixed allocation

- Clean, unbiased estimate of impact
- Some users get the worse model longer
- Clear yes or no decisions
- Best for major launches

BANDIT

Adaptive allocation

- Moves traffic towards the winner
- Lower cost of exploration
- Effect estimates are harder to interpret
- Best for many short lived options

VERDICT Use A/B tests when you need trustworthy causal evidence. Use bandits when optimising quickly among many options matters more.

Online evaluation pitfalls

PITFALL	BETTER PRACTICE
⚠ Stopping the test the moment it looks significant.	✓ Fix duration or use sequential methods.
⚠ Ignoring a sample ratio mismatch.	✓ Investigate assignment bugs before reading results.
⚠ Shipping on a primary win with broken guardrails.	✓ Let guardrails veto harmful launches.
⚠ Treating unplanned segment wins as proof.	✓ Confirm exploratory findings with a new test.

Knowledge check

? A new model lifts click through rate by 3 percent in week one, but the gain shrinks to zero by week three. What happened?

MODEL ANSWER

Most likely a novelty effect: users explored the changed recommendations at first, then returned to normal behaviour. Judge the model on the stable period, consider longer tests, and look at deeper outcomes such as purchases and retention.

WHY IT MATTERS

Short tests can mistake curiosity for lasting value.

Section recap

KEY TAKEAWAYS

- Online tests measure real behaviour and value
- Randomise users and pre register the analysis
- Use one primary metric plus guardrails
- Plan sample size and run full weekly cycles
- Guard against peeking, ratio mismatch and novelty
- Report effects with intervals, not just p values

TRY THIS

Write a one page experiment plan for your next model launch, including hypothesis, metrics, sample size and decision rules.

NEXT UP

Section 19: evaluation in production.

SECTION 19

19

Evaluation in production

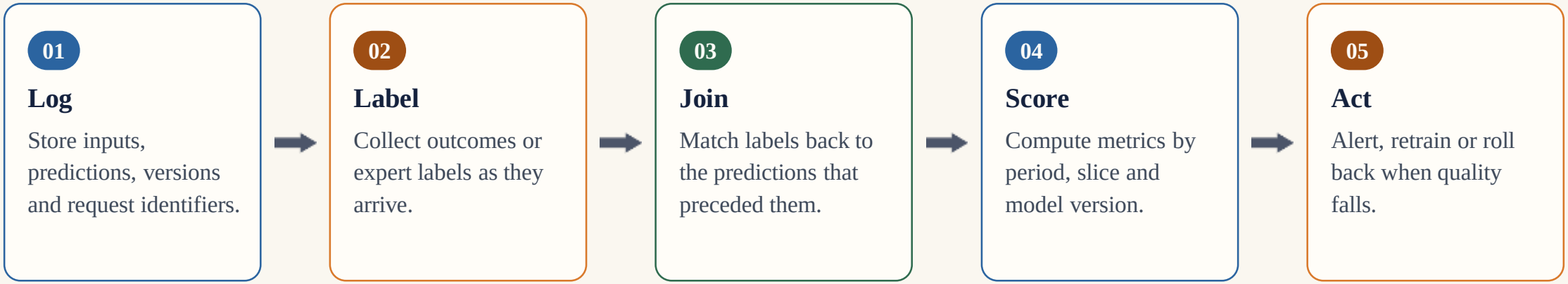
Evaluation does not stop at launch. It becomes a continuous process.

IN THIS SECTION

- Continuous evaluation
- Delayed and partial labels
- Drift as an early warning
- Evaluation gates in pipelines
- Model reports

SECTION 19 · PRODUCTION

The continuous evaluation loop



The same metric code used offline should run in production, so numbers are comparable across the model’s whole life.

SECTION 19 · PRODUCTION

Delayed and partial labels

Outcomes like defaults, churn or diagnoses can take months, and you only see outcomes for cases the model allowed through.

Proxies

Track early signals such as payment delays or user complaints.

Sampled labelling

Have experts label a random sample weekly to estimate accuracy.

Selection bias

Keep a small random holdout that bypasses the model to see unbiased outcomes.

SECTION 19 · PRODUCTION

Drift as an early warning

When labels are slow, shifts in inputs and predictions are the first sign that evaluation results may no longer hold.

Input drift

Changes in feature distributions compared with the training reference.

Prediction drift

Changes in the share of positives or the score distribution.

Treat as a prompt

Drift calls for investigation and fresh evaluation, not automatic panic.

Measuring input drift with the population stability index

Age band	Reference vs current share	Ref	Now	Contribution
18–25		10%	4%	0.055
26–35		20%	12%	0.041
36–45		30%	26%	0.006
46–60		25%	32%	0.017
60+		15%	26%	0.060

Training reference This month

PSI = 0.179

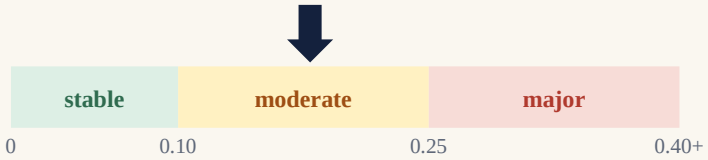
The two biggest contributions come from the extremes: younger customers fell, over-60s grew.

FORMULA

PSI = Σ (now − ref) × ln(now ÷ ref)

60+ band: (0.26 − 0.15) × ln(0.26 ÷ 0.15) = 0.11 × 0.55 = 0.060

COMMON RULE OF THUMB



NEXT STEP

Drift is a prompt, not a verdict. Check accuracy on the newest labelled data, then decide whether to retrain.

Evaluation gates in an ML pipeline

Gate	Check	Example rule
Data	Schema, ranges, freshness	No new nulls in key fields
Quality	Primary metric versus champion	Not worse beyond the interval
Slices	Key segment metrics	No segment drops beyond tolerance
Fairness	Group metric gaps	Within the agreed limit
Calibration	Reliability on recent data	Calibration error under target
Behaviour	Behavioural test suite	All critical tests pass

A standard model evaluation report

Summary	Version, data window and headline results with intervals.
Comparison	Champion versus challenger on all key metrics.
Slices	Per segment results with counts.
Responsible AI	Fairness, calibration and robustness checks.
Errors	Top error categories and example cases.
Decision	Recommendation, approver and date.



Make it routine

Generate the same report automatically for every candidate and every production period, so comparisons are effortless and consistent.

SECTION 19 · PRODUCTION

Production evaluation pitfalls

PITFALL	BETTER PRACTICE
⚠ No logged predictions to evaluate later.	→ ✓ Log predictions with versions and identifiers.
⚠ Only measuring outcomes the model allowed.	→ ✓ Keep a small random holdout for unbiased labels.
⚠ Different metric code offline and online.	→ ✓ Share one evaluation library everywhere.
⚠ Reports produced by hand occasionally.	→ ✓ Automate reports on a fixed schedule.

SECTION 19 · PRODUCTION

Section recap

KEY TAKEAWAYS

- Evaluation continues for the model's whole life
- Join delayed labels back to logged predictions
- Use proxies and sampled labels meanwhile
- Treat drift as an early warning
- Gate releases with automated checks
- Standardise and automate evaluation reports

TRY THIS

Design a weekly evaluation report for a live model: data sources, metrics, slices, owners and the actions triggered.

NEXT UP

Section 20: an end to end case study.

SECTION 20

20

End to end case study

Evaluating a fictional hospital readmission risk model from first question to live use.

IN THIS SECTION

- Framing the problem
- Splits and baselines
- Metrics and thresholds
- Fairness and calibration
- Online evaluation
- Lessons

SECTION 20 · CASE STUDY

The scenario

A fictional hospital wants to identify patients at high risk of readmission within 30 days so nurses can arrange extra follow up.

Decision

Patients above a threshold receive a follow up phone call and care review.

Capacity

Nurses can manage about 40 extra follow ups per day.

Stakes

Missed high risk patients may return sicker. Unneeded calls cost staff time.

SECTION 20 · CASE STUDY

Evaluation decisions at framing

Question	Decision
Split	Train on earlier years, test on the most recent year, grouped by patient
Baselines	Current nurse checklist and a simple logistic regression
Primary metric	Recall among the top 40 daily patients
Secondary	Precision at 40, average precision, calibration
Fairness	Recall and calibration across age, sex and deprivation bands
Online	Stepped rollout comparing wards, with readmission rate as outcome

SECTION 20 · CASE STUDY

Offline evaluation results path

01
Leakage audit
Removed a discharge code entered after readmission.

02
Baselines
Checklist and logistic regression scored first.

03
Candidates
Gradient boosting compared with intervals across folds.

04
Slices
Weaker recall found for patients over 85.

05
Calibration
Isotonic calibration applied on validation data.

The leakage audit removed the single most powerful feature. The headline metric dropped, but the model became honest.

SECTION 20 · CASE STUDY

Choosing the threshold



Rather than a probability cut off, the team ranked patients daily and flagged the top 40, matching nurse capacity exactly.

Capacity first

Evaluation used recall and precision at 40 per day, the real operating point.

Cost view

Estimated nurse hours against avoided readmissions for each candidate model.

Calibrated scores

Calibrated risks let clinicians read scores as meaningful percentages.

One day's top 40 queue, the way the nurses see it

MODEL'S TOP 40 OF 300 DISCHARGES · RANKED BY RISK

#1 readmitted	#2 readmitted	#3 fine	#4 readmitted	#5 readmitted	#6 fine	#7 readmitted	#8 fine
#9 readmitted	#10 fine	#11 readmitted	#12 fine	#13 fine	#14 readmitted	#15 fine	#16 fine
#17 readmitted	#18 fine	#19 fine	#20 readmitted	#21 fine	#22 fine	#23 fine	#24 readmitted
#25 fine	#26 fine	#27 fine	#28 fine	#29 readmitted	#30 fine	#31 fine	#32 fine
#33 readmitted	#34 fine	#35 fine	#36 fine	#37 fine	#38 readmitted	#39 fine	#40 fine

Hits cluster near the top: the ranking works best where nurses start calling.

Readmitted within 30 days (14)

Not readmitted (26)

PRECISION AT 40

35%

14 ÷ 40 calls

RECALL AT 40

39%

14 ÷ 36 readmissions

LIFT

2.9×

35% ÷ 12% base rate

SAME CAPACITY, THREE RANKINGS

	Checklist	Model	Perfect
Readmitted in top 40	9	**14**	36
Precision at 40	23%	**35%**	90%
Recall at 40	25%	**39%**	100%
Lift over random	1.9×	**2.9×**	7.5×

! Report against capacity and against the ceiling: a perfect ranker would reach 90% precision, not 100%, because only 36 of 40 calls can be hits.

SECTION 20 · CASE STUDY

Fairness findings



Slicing showed patients over 85 had lower recall because many lived in care homes, which the data captured poorly.

Root cause

A measurement gap: care home status was missing for many records.

Action

Added a reliable residence feature and checked recall again with intervals.

Governance

Documented remaining gaps and set monitoring on age band recall.

SECTION 20 · CASE STUDY

Online evaluation plan

1. **Shadow.** Scored patients silently for a month and compared with the checklist.
2. **Clinician review.** Nurses reviewed a sample of flags to check clinical sense.
3. **Stepped rollout.** Wards adopted the model in stages for comparison.
4. **Outcomes.** Tracked 30 day readmissions, call completion and workload.
5. **Guardrails.** Watched missed high risk patients and nurse overtime.
6. **Monthly report.** Automated slices, calibration and drift for governance review.

SECTION 20 · CASE STUDY

Lessons from the case study

The most valuable evaluation work was not choosing the algorithm. It was removing leakage, matching capacity and finding who was underserved.

Honesty first

A lower but honest score beats a higher leaked one.

Operating point

Metrics at real capacity made results meaningful to nurses.

Slices matter

Averages hid the weakest group, the oldest patients.

Section recap

KEY TAKEAWAYS

- Frame the decision, capacity and stakes first
- Audit leakage before trusting any metric
- Evaluate at the real operating point
- Calibrate scores people will read as risks
- Slice for fairness and fix measurement gaps
- Confirm value with careful online evaluation

TRY THIS

Rewrite the framing table for a model in your own domain, including split, baselines, metrics, fairness and online plan.

NEXT UP

Section 21: checklists, glossary and final quiz.

SECTION 21

21

Checklists, glossary and next steps

Reusable tools for every evaluation you run from now on.

IN THIS SECTION

- Offline evaluation checklist
- Responsible AI checklist
- Launch and online checklist
- Anti patterns
- Glossary
- Final knowledge check

SECTION 21 · TOOLKIT

Offline evaluation checklist

- [ ] Business goal and error costs written down
- [ ] Split respects time, groups and duplicates
- [ ] Leakage audit completed
- [ ] Simple baselines evaluated
- [ ] Metrics chosen before experiments
- [ ] Threshold chosen on validation data
- [ ] Confidence intervals reported
- [ ] Several seeds or folds compared
- [ ] Test set opened only once
- [ ] Results reproducible from code

SECTION 21 · TOOLKIT

Responsible AI checklist

- [ ] Slices by key segments reported
- [ ] Fairness metrics chosen and justified
- [ ] Intersectional groups examined
- [ ] Calibration checked overall and by group
- [ ] Robustness to realistic shifts tested
- [ ] Out of distribution handling defined
- [ ] Explanations reviewed by experts
- [ ] Limitations written in the model card
- [ ] Sign off from risk or legal where needed
- [ ] Monitoring plan covers group metrics

SECTION 21 · TOOLKIT

Launch and online checklist

- [ ] Shadow or offline replay completed
- [ ] Experiment hypothesis written
- [ ] Primary, secondary and guardrail metrics set
- [ ] Sample size and duration calculated
- [ ] Randomisation unit chosen
- [ ] Sample ratio checks automated
- [ ] Stopping rules agreed in advance
- [ ] Predictions logged with versions
- [ ] Label collection and joining planned
- [ ] Evaluation report scheduled

SECTION 21 · TOOLKIT

Evaluation anti patterns

PITFALL	BETTER PRACTICE
⚠ The leaderboard chaser: tuning until the test set agrees.	→ ✓ Keep a locked test set and fresh confirmation data.
⚠ The single number: one metric, no interval, no baseline.	→ ✓ Report several metrics, baselines and uncertainty.
⚠ The average illusion: no slices, no fairness checks.	→ ✓ Slice results and measure group gaps.
⚠ The launch and forget: no evaluation after release.	→ ✓ Evaluate continuously with logged predictions.

The whole evaluation journey, with its four traps



SECTION 21 · TOOLKIT

Glossary, part one

Term	Meaning
Baseline	A simple reference model or rule the model must beat
Bootstrap	Resampling the test set to estimate uncertainty
Calibration	Agreement between predicted probabilities and observed frequencies
Confusion matrix	Table of true and false positives and negatives
Cross validation	Rotating validation folds for stable estimates
Data leakage	Information unavailable at prediction time used in evaluation
F1 score	A balance of precision and recall in one number
Generalisation	Performance on new, unseen data

SECTION 21 · TOOLKIT

Glossary, part two

Term	Meaning
Guardrail metric	A metric that must not worsen during an experiment
NDCG	Ranking metric rewarding relevant items placed near the top
Precision	Share of flagged items that are truly positive
Recall	Share of true positives the model found
ROC AUC	Probability a positive is ranked above a negative
Slice	A subgroup of data evaluated separately
Stratification	Keeping class or group proportions consistent across splits
Test set	Data held back for one final, honest estimate

SECTION 21 · TOOLKIT

Final knowledge check 1

? Your model beats the champion by 0.4 points of AUC on one test run. What must you check before claiming success?

MODEL ANSWER

Compute confidence intervals and a paired comparison on the same items, run several seeds, confirm no tuning happened on the test set, and check that slices and calibration did not worsen. Then judge whether the gain is large enough to matter for the business.

WHY IT MATTERS

Small differences are often noise until proven otherwise.

SECTION 21 · TOOLKIT

Final knowledge check 2

? A fraud model is evaluated with a random split and scores far better than in production. What is the likely problem?

MODEL ANSWER

The random split probably mixed time periods and possibly the same customers across training and test, leaking future patterns and identities. Evaluate again with a time based, group aware split and audit features for target and temporal leakage.

WHY IT MATTERS

Split design often matters more than model choice.

SECTION 21 · TOOLKIT

Final knowledge check 3

? An LLM assistant scores well with an LLM judge but users complain answers are wrong. What should you do?

MODEL ANSWER

Validate the judge against expert human ratings, add factual correctness and faithfulness checks to the rubric, expand the golden set with real user failures, and score retrieval separately if the system uses RAG.

WHY IT MATTERS

An unvalidated judge can confidently measure the wrong thing.

SECTION 21 · TOOLKIT

Your learning path from here

01 — Practise
Run the offline checklist on one existing model.

02 — Quantify
Add intervals, seeds and paired tests.

03 — Deepen
Add slices, fairness and robustness checks.

04 — Go live
Design and run a careful A/B test.

05 — Automate
Build evaluation reports into your pipeline.

Recommended reading: Trustworthy Online Controlled Experiments by Kohavi, Tang and Xu, and the Fairlearn and sklearn model evaluation guides.

Good evaluation turns a promising model into a trustworthy one.

☆ Thank you for completing this tutorial. Measure honestly, question surprising results, and let evidence decide what ships.